Deep Orthogonal Decomposition for surrogate modelling of time dependent PDEs

written and submitted by Dawid Kotowski

A thesis presented for the degree of
Master of Science

Supervisor: Prof. Dr. Mario Ohlberger
Faculty of Mathematics and Computer Science
University Münster
Germany, DE

submitted on August 11, 2025

Contents

Todo list

0.1 Notation

We set $\mathbb{N} := \{1, 2, \dots\}$, $\mathbb{N}_0 := \{0, 1, \dots\}$ and $\mathbb{N}_{\geq n} := \{n, n+1, \dots\}$ for $n \in \mathbb{N}$. Next the set $\mathbb{R}^+ := \{x \in \mathbb{R} \,|\, x \geq 0\}$ will define all real numbers, which are non-negative. Furthermore for any set A, we will write $|A|$ to denote its cardinality. If $x \in \mathbb{R}$, the gaussian ceiling function is given via $\lceil x \rceil := \min\{k \in \mathbb{Z} \,:\, k \geq x\}$.

Further if $d \in \mathbb{N}$ and $\|\cdot\|$ is a norm, then for $x \in \mathbb{R}^d$ and $r > 0$ we set $B_{r,\|\cdot\|}(x)$ as the open ball around x with radius r. Additionally $|\cdot|$ denotes the regular euclidean norm on $\mathbb{R}^d$. We usually do not differ between vectors and scalars in ubiquitous notation.

The transpose of a matrix $A \in \mathbb{R}^{d_1 \times d_2}$ for some $d_1, d_2 \in \mathbb{N}$ is given by $A^T \in \mathbb{R}^{d_2 \times d_1}$.

Let $(a_n)_{n \in \mathbb{N}} \in \mathbb{R}$ be some sequence, then we say $(a_n)_n$ has *(convergence of) order* n, if there is $c \in \mathbb{R}$ independent of n, s.t. $a_n = c \cdot n$ for all $n \in \mathbb{N}$. We then denote $a_n = \mathcal{O}(n)$.

Moreover, we define the *multiindex* notation as follows: For $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d$, we write $|\alpha| := \alpha_1 + \dots + \alpha_d$ and $\alpha! := \alpha_1! \cdots \alpha_d!$. If $x \in \mathbb{R}^d$, then we set

$$x^\alpha := \prod_{i=1}^{d} x_i^{\alpha_i}.$$

0.1.1 Partial Differential Equations

For a function $f : X \to Y$ we denote $\operatorname{supp}(f) := \{x \in X \,|\, f(x) \neq 0\}$ as the *support* of f. Let $g : Y \to Z$ be another function, then $g \circ f : X \to Z$ is their *composition.* Setting the standard topology of $\mathbb{R}^d$ lets us define $\overline{A}$ for some $A \subset \mathbb{R}^d$ as the *closure* of A, and ∂A its *boundary.* If A is non-empty, we set $\operatorname{diam}(A) := \sup_{x,y \in A} |x - y|$ as the diameter of A.

Let $\Omega \subset \mathbb{R}^d$ be open. For $n \in \mathbb{N}_0 \cup \{\infty\}$, we denote by $C^n(\Omega)$ the set of n-times continuously differentiable functions on Ω.

For $f \in C^n(\Omega)$ and $\alpha \in \mathbb{N}_0^d$ with $|\alpha| \leq n$ we write

$$D^\alpha f := \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \cdots \partial x_d^{\alpha_d}}.$$

A function $f : \Omega \to \mathbb{R}^m$ for $\Omega \subset \mathbb{R}^d$ is called *Lipschitz continuous*, if there is a constant $L \geq 0$ such that

$$|\, f(x) - f(y) \,| \leq L|x - y|.$$

We further denote with

$$L^p(\Omega) := \left\{ f : \Omega \to \mathbb{R} \,|\, f \text{ is measurable, and } \int_\Omega |f(x)|^p d\mu(x) < \infty \right\}$$

the Lebesgue space and L^p_{loc} as the space of $L^p(\Omega)$ functions with local support. If not specified otherwise we write $\|\cdot\|_p := \|\cdot\|_{L^p(\Omega)}$ for all $1 \leq p \leq \infty$. Note that in critical situations we tend to be explicit about the norms.

0.1.2 Neural Networks

Consider $d_1, d_2 \in \mathbb{N}$ and a matrix $A \in \mathbb{R}^{d_1 \times d_2}$, then the number of non-zero entries of A is given by

$$\|\, A \,\|_{\ell^0} := |\{(i, j) \,:\, A_{i,j} \neq 0\}|.$$

Let $B \in \mathbb{R}^{d_1 \times d_3}$ be some other matrix for $d_3 \in \mathbb{N}$. Then we write

$$\begin{bmatrix} A & B \end{bmatrix} \in \mathbb{R}^{d_1 \times (d_2 + d_3)} \quad \text{or} \quad \left[A \,\middle|\, B \right] \in \mathbb{R}^{d_1 \times (d_2 + d_3)},$$

for the stacking of matrices, where the latter notation is meant to symbolize a strong delineation between blocks. Similar applies to vertical stacking, if $A \in \mathbb{R}^{d_1 \times d_2}$ and $B \in \mathbb{R}^{d_3 \times d_2}$. As a special case, this applies to vectors as well.

Chapter 1

Introduction

Write an Introduction

1.1 Time Dependent PDEs

The study and analysis of partial differential equations (PDEs) started with the works of Euler, d'Alembert, Lagrange and Laplace (see [BB98] for this introduction). As so often in mathematics it started off with a set of simple problems, such as the vibrating string in one dimension as modelled by d'Alembert (1752)

$$\frac{\partial^2}{\partial t^2}u(x,t) = \frac{\partial^2}{\partial x^2}u(x,t), \quad \text{for } u \in C^2(\mathbb{R} \times \mathbb{R}^+), x \in \mathbb{R} \text{ and } t \in \mathbb{R}^+,$$

the extension to three dimensions by Euler (1759) and Bernoulli (1762), the Laplace equation for gravitational fields (1780)

$$\Delta u(x) = 0, \quad \text{for } u \in C^2(\mathbb{R}^3) \text{ and } x \in \mathbb{R}^3,$$

and finally the heat equation, as introduced by Fourier (1810 - 1822)

$$\frac{\partial}{\partial t}u(x,t) = \Delta u(x,t), \quad \text{for } u \in C^2(\mathbb{R}^3 \times \mathbb{R}^+), x \in \mathbb{R}^3 \text{ and } t \in \mathbb{R}^+.$$

As any posed problem calls for a solution to it, there have been simultaneous developments with regard to the calculation methods; the method of seperation of variables was given by Euler and d'Alembert for the study of spherical harmonics and Fourier introduced similar ideas for the heat equation. For the Laplace equation, the Green functions were introduced in 1835. Green also discovered an important principle for solvability of the (later so called) Dirichlet problem

$$\begin{cases} \Delta u(x) = 0, & \text{for } x \in \Omega \subset \mathbb{R}^3, \\ u(x) = g(x), & \text{for } x \in \partial\Omega, \end{cases}$$

for some function $g \in C^0(\mathbb{R}^3)$, which revolved around minimizing an energy functional $E : C^2(\mathbb{R}^3) \to \mathbb{R}^+$, thus founding the variational principle.

Though these were first attempts at solving such problems, the idea of Green in particular led to a few analytical problems, by the lack of mathematical rigor. Together with the contribution of Cauchy on the *initial value* problem, this laid the groundwork for the field of *Existence Theory*, which tries to establish existence results for PDEs.

Notable here are the alternating method of Schwarz (1870) using the idea of domain decomposition and especially Neumanns method of integral equations developed in 1877, which would go on to be the starting point for the weak form and subsequently the Sobolev spaces. These were a necessity after B. Levi showed in 1906, that a general minimizing sequence for the Dirichlet integral was a Cauchy sequence and would therefore converge in the right space, which was then given due to the Sobolev embedding theorems.

To tackle the problem arising from "exotic" examples and the complicated nature of some PDEs, Hadamard proposed a classification of the most common linear PDEs using its characteristic polynomial, which is given by replacing all partial derivatives $\partial/\partial x_j$ by the algebraic variable ξ_j. In that case the PDE is called

1. *elliptic*, if after a change of varibles the polynomial can be written as $\xi_1^2 + \xi_2^2 + \cdots + \xi_n^2$,
2. *parabolic*, if after a change of varibles the polynomial can be written as $\xi_1 + \xi_2^2 + \cdots + \xi_n^2$,
3. *hyperbolic*, if after a change of varibles the polynomial can be written as $\xi_1^2 - (\xi_2^2 + \cdots + \xi_n^2)$.

These analytical results can be expanded a lot more, but what is most interesting in terms of this paper, are the Sobolev spaces. They, and the closely connected Bochner spaces, are the relevant solution spaces to the limits of all, in some sense, consistent numerical schemes, since the solutions are at least embedded in them. This follows quite natural, as any numerical scheme is basically bounded by the necessity of being ran on a computer and thus having a finite dimensional space it defines its approximation on, and at the very least the Sobolev and Bochner spaces admit any piecewise smooth function.

1.2 Neural networks

1.3 Motivation

1.4 Literature review

1.5 Outline

Chapter 2

Theoretical Background

2.1 Parametric Partial Differential Equations

2.1.1 Sobolev Spaces

For our work with Partial Differential Equation, and subsequent error approximations, we want to introduce Sobolev spaces. These are generalized differentiable spaces, where a weak derivative is given, as an analogon to the classic derivative in the $C^n(\Omega)$ space. They are employed in order to exploit their compact embedding in the Hölder spaces, which are generalizations of $C^n(\Omega)$. We will not go in much detail about the theory of Sobolev spaces, since we mostly concern ourselves with finite dimensional subspaces, but some ground truth must be discussed.

For this subsection we primary use [Alt16], and for the first upcoming definition [Dac04].

Definition 2.1 (Lipschitz Domain)**.** Let $d \in \mathbb{N}$, and $\Omega \subset \mathbb{R}^d$. Then Ω is called a *Lipschitz domain* if for every $x \in \partial\Omega$ and some $r > 0$, there is $U := B_{r,|\cdot|}(x) \subset \mathbb{R}^d$ and an isomorphic map $H : Q \to B_{r,|\cdot|}(x)$ with $Q := \{x \in \mathbb{R}^d \,|\, \| x \|_{\ell^1} < d\}$ such that

$$H \in C^k(\overline{Q}), \quad H^{-1} \in C^k(\overline{U}), \quad H(Q_+) = U \cap \Omega, \quad H(Q_0) = U \cap \partial\Omega$$

where $Q_+ := \{x \in Q \,|\, x_d > 0\}$ and similarily $Q_0 := \{x \in Q \,|\, x_d = 0\}$, and H is only in $C^{0,1}$.

From now on, let $d \in \mathbb{N}$ and, if not given otherwise, $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain.

Definition 2.2 (Weak Derivative)**.** Let $\alpha \in \mathbb{N}_0^d$ be a multiindex. A function $u \in L^1_{\text{loc}}(\Omega)$ is said to have an *α-th weak derivative*, if there is a function $v_\alpha \in L^1_{\text{loc}}(\Omega)$ such that

$$\int_\Omega u D^\alpha \varphi = (-1)^{|\alpha|} \int_\Omega v_\alpha \varphi \quad \forall \varphi \in C_0^\infty(\Omega).$$

We then write $v_\alpha = D^\alpha u$.

Definition 2.3 (Sobolev Space)**.** Let $n \in \mathbb{N}_0$ and $1 \leq p \leq \infty$. Then we define the *Sobolev space* as

$$W^{n,p}(\Omega) := \{f \in L^p(\Omega) \,|\, \| f \|_{W^{n,p}(\Omega)} < \infty\},$$

where the norm on this space is given via

$$\| f \|_{W^{n,p}(\Omega)} := \begin{cases} \left(\sum_{|\alpha| \leq n} \| D^\alpha f \|^p_{L^p(\Omega)} \right)^{1/p}, & \text{if } p < \infty \\ \max_{|\alpha| \leq n} \| D^\alpha f \|_{L^\infty(\Omega)}, & \text{if } p = \infty, \end{cases}$$

for any $f \in W^{n,p}(\Omega)$.

It is possible to extend this definition for $0 < n < 1$ to that of the *Sobolev-Slobodeckij spaces*. In the analysis of partial differential equations these are used to characterize the regularity of Sobolev functions, which are restricted to the boundary $\partial\Omega$. This restriction itself is called a trace operator. One of the statements of relevance under the following definition is this:

If $\Omega \subset \mathbb{R}^d$ is a regular open set (for example, if Ω is a Lipschitz domain), and if $u \in W^{1,p}(\Omega)$, then in fact $u \mapsto u|_{\partial\Omega}$ as a trace operator is surjective from $W^{1,p}(\Omega)$ onto $W^{1-(1/p),p}(\partial\Omega)$ and

$$\| u|_{\partial\Omega} \|_{W^{1-(1/p),p}(\partial\Omega)} \leq C \| u \|_{W^{1,p}(\Omega)},$$

for some $C > 0$ independent of p and u. This of course only makes sense in light of the next definition.

Definition 2.4 (Sobolev-Slobodeckij Space)**.** Let $0 < s < 1$ and $1 \leq p \leq \infty$. Then we define the *Sobolev-Slobodeckij space* as

$$W^{s,p}(\Omega) := \{ f \in L^p(\Omega) \mid \| f \|_{W^{s,p}(\Omega)} < \infty \},$$

where the norm on this space is given via

$$\| f \|_{W^{s,p}(\Omega)} := \begin{cases} \left(\| f \|^p_{L^p(\Omega)} + \displaystyle\int_\Omega \int_\Omega \left(\frac{|f(x) - f(y)|}{|x-y|^{s+d/p}} \right)^p dxdy \right)^{1/p}, & \text{if } p < \infty \\ \max \left\{ \| f \|_{L^\infty(\Omega)}, \operatorname*{ess\,sup}_{x,y \in \Omega} \dfrac{|f(x) - f(y)|}{|x-y|^s} \right\}, & \text{if } p = \infty, \end{cases}$$

for any $f \in W^{s,p}(\Omega)$.

2.1.2 Existence and Uniqueness

The following subsection is meant as a summary of any literature concerning the existence of actual solutions for a subclass of time-dependent partial differential equations, to give the reader a feeling for the current developments in the field. Note here, that this is of course not general for all time-dependend problems since the existence of a solution to equations such as the incompressible Navier-Stokes equation are still a pending research question.

Theorem 2.5 (Hille-Yosida)**.** *Let A be a maximal monotone operator. Then, given any $u_0 \in D(A)$ there is a unique function*

$$u \in C^1([0, \infty]; H^1_0(\Omega))$$

2.1.3 Reduced Order Modelling

Introduce the general notion of the POD

Introduce KnW and its relation to the continous case

For this subsection we define $u(\mu_j, t_k) \in \mathbb{R}^{N_h}$ for $\mu_j \in \mathcal{P}$ and $t_k \in \mathcal{T}$ to be some vectors, later representing the solutions to a parametric PDE for a given parameter and time point in a given discrete subspace of the solution manifold, sampled uniformly and iid from their respective spaces. Here $\mathcal{P}$ represents and arbitrary compact subspace of $\mathbb{R}^{p+q}$, where $p, q > 0$ and $\mathcal{T} = [0, T]$ for some

$T > 0$. Further we let $I \subset \mathbb{N}$ define some index set. Additionally we let $N_s := |I|$ and N_t denote the number of time samples t_k.

Here we will discuss prerequisites to the Proper Orthogonal Decomposition and related concepts, such as the Kolmogorov n-width, which is closely related to the partial sums of the eigenvalues of the POD matrix. This is among the results, this section will cover. We base this chapter on [QMN15].

Definition 2.6 (Proper Orthogonal Decomposition)**.** Assuming $N_{data} = N_s N_t < \infty$, we define by $U_j = [u(\mu_j, t_k)]_{k=1}^{N_t} \in \mathbb{R}^{N_h \times N_t}$ the snapshot matrix for a given $j \in I$ and consequently $U = [U_j]_{j \in I} \in \mathbb{R}^{N_h \times N_{data}}$ the collective snapshot matrix. Then the (discrete) correlation matrix is given by

$$K = \frac{|\mathcal{P} \times \mathcal{T}|}{N_{data}} U U^T \in \mathbb{R}^{N_h \times N_h}. \tag{2.1}$$

K is symmetric and semi-positive definite and thus has real and positive eigenvalues $\sigma_1^2 \geq \cdots \geq \sigma_r^2$. We further call ψ_k the eigenvectors of the correlation matrix K. With these, the corresponding POD basis of dimension $N < N_h$ is given via

$$\xi_k = \frac{1}{\sigma_k} K \psi_k, \quad 1 \leq k \leq N, \tag{2.2}$$

for the N largest eigenvalues and their corresponding eigenvectors. Further, if $N_s, N_t = \infty$, we define the continuous counterpart to the correlation matrix via

$$K_\infty = \int_{\mathcal{P} \times \mathcal{T}} u(\mu, t) u^T(\mu, t) d(\mu, t) \in \mathbb{R}^{N_h \times N_h}, \tag{2.3}$$

and its real eigenvalues by $\sigma_{k,\infty}^2$.

Proposition 2.7. *Let K and K_∞ and their respective eigenvalues be defined as in (2.6), then letting $N_s, N_t \to \infty$, we get*

$$\sum_{k>N} \sigma_k^2 \longrightarrow \sum_{k>N} \sigma_{k,\infty}^2, \qquad \text{a.s.}$$

Proof.

Insert adequate consideration.

By some consideration there exists a matrix $V_\infty \in \mathbb{R}^{N_h \times N_h}$ such that

$$\begin{aligned} \sum_{k>N} \sigma_{k,\infty}^2 &= \int_{\mathcal{P} \times \mathcal{T}} \| u(\mu, t) - V_\infty V_\infty^T u(\mu, t) \|^2 d(\mu, t) \\ &= \min_{W \in \mathbb{R}^{N_h \times N} : W^T W = I} \int_{\mu \times \mathcal{T}} \| u(\mu, t) - W W^T u(\mu, t) \|^2 d(\mu, t). \end{aligned}$$

Thus, since this is the optimal choice of an approximation matrix for the continuous case and $V \in \mathbb{R}^{N_h \times N}$ is only optimal in the discrete setting and not necessarily the continuous case, we have

$$\sum_{k>N} \sigma_{k,\infty}^2 \leq \int_{\mathcal{P} \times \mathcal{T}} \| u(\mu, t) - V V^T u(\mu, t) \|^2 d(\mu, t) = \sum_{k>N} \sigma_k^2. \tag{2.4}$$

To properly handle the notion of $N_s, N_t \to \infty$ in a regularized fashion, we introduce some arbitrary samples $I_{train} \subset I$ and $\mathcal{T}_{train} \subset \mathcal{T}$, such that I_{train} and $\mathcal{T}_{train}$ are finite. Note, that in general this does rely on the axiom of choice in the theoretical setting, but since the test samples are already known in practice, this is rather an unnecessary remark. We then write entry-wise for $m, l = 1, \dots N_h$

$$[K_\infty - K]_{ml} = \int_{\mathcal{P}\times\mathcal{T}} [u(\mu,t)]_m [u(\mu,t)]_l d(\mu,t) - \frac{|\mathcal{P}\times\mathcal{T}|}{N_{data}} \sum_{j\in I_{train}} \sum_{k\in\mathcal{T}_{train}} [u(\mu_j,t_k)]_m [u(\mu_j,t_k)]_l.$$

By the uniform iid sampling assumed in the introduction to Section (2.1), we know that with the entry-wise boundedness of the PDE solutions, the resulting matrix is subject to the strong law of large numbers, and thus converges to zero, given the entry-wise matrix norm $\|\cdot\|_1$ as $N_t, N_s \to \infty$ almost surely.

Why is the boundedness clear here? Independent of the specific PDE?

Now using Bauer-Fike's Theorem with the 1-norm, we have upon ordering, that for each $\sigma_{k,\infty}^2$ there is a σ_k^2 in the spectrum of K, such that

$$|\sigma_{k,\infty}^2 - \sigma_k^2| \leq \kappa(V) \| K_\infty - K \|_1,$$

where $\kappa(V)$ denotes the condition number of the eigenvector matrix of K.

Why is this sufficient? Does the condition number always stay bounded even throughout growing Ns, Nt?

Now by convergence of $\| K_\infty - K \| \to 0$ almost surely for $N_s, N_t \to \infty$, we conclude the proof. □

Proposition 2.8. Let $\mathcal{V}_N = \{W \in \mathbb{R}^{N_h\times N} \,|\, W^T W = I_N\}$ be the set of all N-dimensional orthonormal bases, $N_{data} := N_t N_s$ and $V = [\xi]_{i=1}^N$ be the POD matrix, then

$$\frac{|\,\mathcal{P}\times\mathcal{T}\,|}{N_{data}} \sum_{i\in I, k=1,\dots,N_t} \| u(\mu_j,t_k) - VV^T u(\mu_j,t_k) \|_2^2$$
$$= \min_{W\in\mathcal{V}_N} \frac{|\,\mathcal{P}\times\mathcal{T}\,|}{N_{data}} \sum_{i\in I, k=1,\dots,N_t} \| u(\mu_j,t_k) - WW^T u(\mu_j,t_k) \|_2^2 = \sum_{j=N+1}^{r} \sigma_k^2.$$

Definition 2.9 (Linear Kolmogorov N-width)**.** Let $\mathcal{S}_{N_h} = \{u(\mu,t) \in \mathbb{R}^{N_h} \,|\, (\mu,t) \in \mathcal{P}\times\mathcal{T}\}$ be some manifold. The linear Kolmogorov N-width of $\mathcal{S}_{N_h}$ is defined as

$$d_N(S_{N_h}) = \inf_{V\in\mathbb{R}^{N_h\times N}} \sup_{u\in S_{N_h}} \| u - VV^T u \|.$$

Definition 2.10 (Nonlinear Kolmogorov n-width)**.** The nonlinear Kolmogorov n-width of the reduced manifold $\mathcal{S}_N = \{q(\mu,t) = V^T u(\mu,t) \,|\, (\mu,t) \in \mathcal{P}\times\mathcal{T}\}$ is defined as

$$\delta_n(\mathcal{S}_N) = \inf_{\psi\in C(\mathbb{R}^N,\mathbb{R}^n), \psi'\in C(\mathbb{R}^n,\mathbb{R}^N)} \sup_{u\in S_{N_h}} \| u - \psi(\psi'(u)) \|.$$

2.2 Neural Networks

In this section, we will introduce the general notion of neural networks. Specifically important for us will be concatenations of perceptrons, which are a specific subclass of networks, the so called *feedforward neural network*.

Definition 2.11 (Neural Network)**.** Let $d, L \in \mathbb{N}$, then a *neural network* Φ *with input dimension* d *and* L *layers* is defined via

$$\Phi = ((A_1, b_1), (A_2, b_2), \dots, (A_L, b_L)),$$

such that $N_0 = d$ and $N_1, \dots, N_L \in \mathbb{N}$, where each A_l is an $N_l \times \sum_{k=0}^{l-1} N_k$ dimensional matrix, and $b_l \in \mathbb{R}^{N_l}$.

We further define $R_\rho(\Phi) : \mathbb{R}^d \to \mathbb{R}^{N_L}$ as a *realizations of* Φ *with activation function* ρ for some arbitrary map $\rho : \mathbb{R} \to \mathbb{R}$ as the result of the following scheme for some input $x \in R^d$:

$$\begin{aligned} x_0 &:= x, \\ x_l &:= \rho\left(A_l \left[x_0^T \;\middle|\; \cdots \;\middle|\; x_{l-1}^T\right]^T + b_l\right), \quad \text{for } l = 1, \dots, L-1, \\ x_L &:= A_L \left[x_0^T \;\middle|\; \cdots \;\middle|\; x_{l-1}^T\right]^T + b_L, \end{aligned}$$

where ρ acts componentwise, i.e. $\rho(y) := [\rho(y_1), \dots, \rho(y_m)]^T \in \mathbb{R}^m$ for each $y = [y_1, \dots, y_m] \in \mathbb{R}^m$. Note that we can write

$$A_l = \left[A_{l,x_0} \;\middle|\; \cdots \;\middle|\; A_{l,x_{l-1}}\right],$$

with A_{l,x_k} being a $N_l \times N_k$ dimensional matrix for $k = 0, \dots, l-1$ and $l = 1, \dots, L$. This comes in handy to portray the scheme via

$$\begin{aligned} x_l &:= \rho\left(A_{l,x_0} x_0 + \cdots + A_{l,x_{l-1}} x_{l-1} + b_l\right), \quad \text{for } l = 1, \dots, L-1, \\ x_L &:= A_{L,x_{L-1}} x_{L-1} + b_L. \end{aligned}$$

Moreover call $N_\Phi := d + \sum_{l=1}^{L} N_l$ the *number of neurons* of the network, $L_\Phi := L$ the *number of layers*, and finally $\omega_\Phi := \sum_{l=1}^{L} (\| A_l \|_{\ell^0} + \| b_l \|_{\ell^0})$ the *number of active weights*.

Remark 2.12. As a note, we can rewrite known neural network architectures, such as a convolutional autoencoder, as a neural network according to Definition (2.11).

Motivate this with some example.

From this we can infer a simpler architechture, describing a network which omits any connection of non-neighboring layers.

Definition 2.13 (Standard Neural Network)**.** If a neural network $\Phi = ((A_1, b_1), \dots, (A_L, b_L))$ now has the characteristic

$$A_{l,x_i} = 0, \quad \text{for } l = 1, \dots, L, \text{ and } i = 0, \dots, l-2,$$

then we call Φ a *standard neural network.*

When considering the similarities and differences between two distinct neural networks $\Phi_1 = ((A_1, b_1), \dots, (A_L, b_L))$ and $\Phi_2 = ((\hat{A}_1, \hat{b}_1), \dots, (\hat{A}_L, \hat{b}_L))$, one can easily see, that if $[A_1]_{ij} = 0$, while $[\hat{A}_1]_{ij} \neq 0$ for some i, j, then these networks are severely different, whereas if they only differ in the specific non-zero weight, they could be recovered from each other through training. This inspires the definition of *an architecture of a network.*

Definition 2.14 (Neural Network Architecture)**.** For some $d, L \in \mathbb{N}$, we define a *neural network architecture* $\mathcal{A}$ *with input dimension* d *and* L *layers* to be a binary neural network, i.e. for all entries of $A_i, b_i \in \{0, 1\}$ for $i = 1, \dots, L$ of $\mathcal{A}$.

We further say that a *neural network* $\Phi = ((\hat{A}_1, \hat{b}_1), \dots, (\hat{A}_L, \hat{b}_L))$ has *architecture* $\mathcal{A}$, if

1. $N_l(\Phi) = N_l$ for all $l = 1, \dots, L$, and
2. $[\hat{A}_l]_{ij} \neq 0$ implies $[A_l]_{ij} \neq 0$ for all $l = 1, \dots, L$ with $i = 1, \dots, N_l$ and $j = 1, \dots, \sum_{k=0}^{l-1} N_k$.

Since the associated realization of a neural network $R_\rho(\Phi)$ requires the definition of ρ, the activation function, we will define the most commonly used one in literature. It will help a lot with some of the constructions, since it has a simple nature, but also provides the network with the necessary non-linearity.

Definition 2.15 (ReLU)**.** The function

$$\rho : \mathbb{R} \to \mathbb{R}; \quad x \mapsto \max(0, x),$$

is called *ReLU* (Rectified Linear Unit) *activation function.*

Given such an activation function, Kurt Hornik provided in 1990 the first contribution to a field, which would emerge centuries later: Approximation theory. He proved for any $L^p(\mu)$ function, for μ some finite measure, the existence of an arbitrarily close ReLU valued neural network [Hor91].

Theorem 2.16 (Hornik's Theorem)**.** *Define for some continuous, unbounded and non-constant actiation function ρ the set of all feedforward neural networks, as*

$$\mathcal{N}_d(\rho) := \bigcup_{L=1}^{\infty} \Big\{ R_\rho(\Phi) : \mathbb{R}^d \to \mathbb{R} \,\big|\, \Phi \textit{ is neural network with } L \textit{ layers} \Big\}.$$

Then $\mathcal{N}_d(\rho)$ lies dense in $L^p(\mu)$ for every finite measure μ, $p \in (0, \infty)$.

The proof of this statement leverages on Hahn Banach, but thus does not provide the specific architecture of the required neural network, which would be used for such an approximation. We aim significantly higher, namely at a bound on the necessary non-zero weights and layers for such an approximation, and will thus not mention the proof beyond this outline.

2.2.1 Network concatenations

In this subsection we will discuss the concatenations of multiple neural networks and the effects on the characteristic numbers of it. The complexity of a concatenated network can be given nicely, when the used activation function is

Definition 2.17 (Network Concatenation)**.** Let $L_1, L_2 \in \mathbb{N}$ and

$$\Phi^1 = ((A_1, b_1), \dots, (A_{L_1}, b_{L_1})), \quad \Phi^2 = ((B_1, c_1), \dots, (B_{L_2}, c_{L_2})),$$

be two neural networks such that the input layer of Φ^1 has the same dimension as the output layer of Φ^2. Then, $\Phi^1 \bullet \Phi^2$ denotes the following $L_1 + L_2 - 1$ layered network:

$$\Phi^1 \bullet \Phi^2 := \Big(((B_1, c_1), \dots, (B_{L_2-1}, c_{L_2-1})), ((\widetilde{A}_1, \widetilde{b}_1), \dots, (\widetilde{A}_{L_1}, \widetilde{b}_{L_1})) \Big),$$

where

$$\widetilde{A}_l^1 := \Big[A_{l,x_0} \, B_{L_2} \,\Big|\, A_{l,x_1} \,\Big|\, \cdots \,\Big|\, A_{l,x_{l-1}} \Big], \quad \text{and} \quad \widetilde{b}_l^1 := A_{l,x_0} \, c_{L_2} + b_l,$$

for $l = 1, \dots, L_1$. We call $\Phi^1 \bullet \Phi^2$ the *concatenation* of Φ^1 and Φ^2.

Lemma 2.18 (Neutral Neural Network)**.** Let $\Phi^{\mathrm{Id}} = ((A_1, b_1), (A_2, b_2))$ be a d-dimensional neural network, with

$$A_1 := \begin{bmatrix} \mathrm{Id}_{\mathbb{R}^d} \\ -\mathrm{Id}_{\mathbb{R}^d} \end{bmatrix}, \quad b_1 := 0, \quad A_2 := \left[0_{\mathbb{R}^{d\times d}} \,\middle|\, \mathrm{Id}_{\mathbb{R}^d} \,\middle|\, -\mathrm{Id}_{\mathbb{R}^d} \right], \quad b_2 := 0.$$

Then $R_\rho(\Phi^{\mathrm{Id}}) = \mathrm{Id}_{\mathbb{R}^d}$.

Lemma 2.19. For any two neural networks Φ^1 and Φ^1 according to Definition (2.17), $\Phi^1 \odot \Phi^2 := \Phi^1 \bullet \Phi^{\mathrm{id}} \bullet \Phi^2$ is a neural network with d-dimensional input, $L_1 + L_2$ layers with $\omega_{\Phi^1 \odot \Phi^2} \leq 2\omega_{\Phi^1} + 2\omega_{\Phi^2}$ active weights, $N_{\Phi^1 \odot \Phi^2} \leq 2N_{\Phi^1} + 2N_{\Phi^2}$ neurons. Additionally for all $x \in \mathbb{R}^d$, we have

$$R_\rho(\Phi^1 \odot \Phi^2)(x) = R_\rho(\Phi^1) \circ R_\rho(\Phi^2)(x).$$

Proof.

maybe include proof?

Intuitively speaking the latter network must account for all "missing" links, which a direct concatenation would necessarily be missing; any layer after the "switch" from the first to the latter needs to have a correction term in the weights, additionally attributing all potential connections which the first network would impose.

Let w.l.o.g. $L_2 = 2$, since the pass through the first layers of the combined network, is equivalent to the pass through the first layers of the individual network. Then we can show by induction:

1. Base case: $L_2 = 1$. We argue with $A_{L_1} x = A_{L_1, x_0}\, x$ for x of output dimension of Φ^1, since the input dimension of Φ^1 is the output of Φ^2.

$$\begin{aligned} R_\rho(\Phi^1) \circ R_\rho(\Phi^2)(x_0) &= A_{L_1} \left(B_{L_2, x_0} x_0 + B_{L_2, x_1} x_1 + c_{L_2} \right) + b_{L_1} \\ &= A_{L_1, x_0} B_{L_2, x_0} x_0 + A_{L_1, x_0} B_{L_2, x_1} x_1 + \left(A_{L_1, x_0} c_{L_2} + b_{L_1} \right) \\ &= A_{L_1, x_0} B_{L_2} \left[x_0^T \,\middle|\, x_1^T \right]^T + \left(A_{L_1, x_0} c_{L_2} + b_{L_1} \right) \\ &= R_\rho(\Phi^1 \bullet \Phi^2)(x_0) \end{aligned}$$

2. Induction assumption: Let the statement be true for $L_2 \geq 2$ fixed.

3. Induction step: $L_2 \mapsto L_2 + 1$.

□

2.2.2 Network parallelization

We have shown, that there is a possiblity to simply compose networks, as is possible with functions. To further build upon that idea, we will now introduce stacking in the context of definiting a network $P(\Phi^1, \Phi^2)$ from two networks Φ^1 and Φ^2 such that $R_\rho(P(\Phi^1, \Phi^2))(x) = (R_\rho(\Phi^1)(x), R_\rho(\Phi^2)(x))$ for all $x \in \mathbb{R}^d$.

Definition 2.20 (Network Parallelization)**.** Let $L_1, L_2 \in \mathbb{N}$ s.t. $L_1 \leq L_2$ and let

$$\Phi^1 = ((A_1, b_1), \ldots, (A_{L_1}, b_{L_1})), \quad \Phi^2 = ((B_1, c_1), \ldots, (B_{L_2}, c_{L_2})),$$

be two neural networks with d-dimensional input. We call

$$P(\Phi^1, \Phi^2) := \left((C_1, d_1), \ldots, (C_{L_2}, d_{L_2}) \right),$$

the *parallelization of* Φ^1 *and* Φ^2 with

$$C_l := \left[\begin{array}{c|cc|c|cc} A_{l,x_0} & A_{l,x_1} & 0 & \cdots & A_{l,x_{l-1}} & 0 \\ B_{l,x_0} & 0 & B_{l,x_1} & \cdots & 0 & B_{l,x_{l-1}} \end{array}\right], \qquad d_l := \begin{bmatrix} b_l \\ c_l \end{bmatrix}, \qquad \text{for } 1 \leq l < L_1,$$

$$C_{L_1} := \left[\begin{array}{c|cc|c|cc|c|c|c} B_{l,x_0} & 0 & B_{l,x_1} & \cdots & 0 & B_{l,x_{L_1-1}} & B_{l,x_{L_1}} & \cdots & B_{l,x_{l-1}} \end{array}\right], \qquad d_l := c_l,$$

for $L_1 \leq l < L_2$, and

$$C_{L_2} := \left[\begin{array}{c|cc|c|cc|c|c|c} A_{L_1,x_0} & A_{L_1,x_1} & 0 & \cdots & A_{L_1,x_{L_1-1}} & 0 & 0 & \cdots & 0 \\ B_{L_2,x_0} & 0 & B_{L_2,x_1} & \cdots & 0 & B_{L_2,x_{L_1-1}} & B_{L_2,x_{L_1}} & \cdots & B_{L_2,x_{L_2-1}} \end{array}\right],$$

$$d_{L_2} := \begin{bmatrix} b_{L_1} \\ c_{L_2} \end{bmatrix}.$$

A similar construction is used for the case $L_1 > L_2$.

Lemma 2.21. In the setting of Definition (2.20), we have that $P(\Phi^1, \Phi^2)$ is a neural network with d-dimensional input, $\max\{L_1, L_2\}$ layers, $\omega_{P(\Phi^1,\Phi^2)} = \omega_{\Phi^1} + \omega_{\Phi^2}$ active weights and $N_{P(\Phi^1,\Phi^2)} = N_{\Phi^1} + N_{\Phi^2} - d$ neurons. Additionally, for all $x \in \mathbb{R}^d$, we have that

$$R_\varrho(P(\Phi^1, \Phi^2))(x) = (R_\varrho(\Phi^1)(x), R_\varrho(\Phi^2)(x)).$$

maybe include proof?

2.2.3 Standard Neural Networks

Corollary 2.22. Let ρ be the ReLU activation function. Then there are constants $C_1, C_2 > 0$ such that for any neural network Φ with d-dimensional input and $L = L(\Phi)$ layers, $N = N(\Phi)$ neurons and $M = M(\Phi)$ nonzero weights, there is a standard neural network Φ_{st} with d-dimensional input, L layers, at most $C_1 LN$ neurons and $C_2 \cdot (LN + M)$ nonzero weights such that

$$R_\rho(\Phi)(x) = R_\rho(\Phi_{\text{st}})(x)$$

for all $x \in \mathbb{R}^d$.

2.2.4 Approximation Theory

For all further theoretical results we will assume that ρ, the activation function, is the ReLU as given in Definition (2.15) and furthermore define the set of functions bounded in Sobolev norm as

$$\mathcal{F}_{n,d,p,B} := \{f \in W^{n,p}((0,1)^d) \mid \| f \|_{W^{n,p}((0,1)^d)} \leq B\}.$$

To create a decent outline for this subsection, we will shortly discuss the goal and the steps to be taken; in order to show a bound on the weights and layers needed for an approximating neural network, we need to take into account the interpolation error, we may overcome with sufficient regularity. Say we would have a function f in the Sobolev space $W^{2,2}((0,1)^d)$, then it is apparent, that at least 2 derivatives of f must exists, in a piecewise, embedded sense, which can be leveraged to get a Taylor expansion around selected points.

This procedure will deduce a quantifiable error for the network approximating the initial arbitrary Sobolev function f in Sobolev norm, as long as we can bound the error, which the polynomial approximation produces with regard to the network itself.

This leaves us with two main objectives:

1. Bound the Sobolev function at critical equidistant points, using its Taylor expansion.
2. Relate a fixed error rate of the approximation polynomial and some neural network, to the networks complexity.

These two objectives are pursued by both Yarotsky [Yar17] and Gühring et al [GKP19], even though it is worth noting, that Yarotsky only provides the special case $s = 0, B = 1$, since the proof only shows an estimation with a Taylor expansion regarding the L^∞ norm, missing the crucial step of the interpolation bound to change the norm to the Sobolev expanded counterpart and additionally uses the L^∞ norm to approximate multiplication by a neural network.

Tho we will use the statement of Gühring later on, which is the extension of Yarotsky to any choice of s, p and B, we will only give the idea of the proof, since it follows the same pattern as Yarotsky's proof, but does include a lot of further theoretical setting, which will not be of much use in the actual results presented in this paper.

Proposition 2.23. There are constants $c_1, c_2, c_3 > 0$, such that for all $\varepsilon \in (0,1)$ there is a neural network $\Phi_\varepsilon^{\mathrm{sq}}$ with at most $c_1 \cdot \ln(1/\varepsilon)$ nonzero weights, at most $c_2 \cdot \ln(1/\varepsilon)$ layers, at most $c_3 \cdot \ln(1/\varepsilon)$ neurons, and with a 1-dimensional input and output such that for $f(x) := x^2$

$$\| R_\rho(\Phi_\varepsilon^{\mathrm{sq}}) - f \|_{W^{0,\infty}((0,1)^1)} \leq \varepsilon,$$

and $R_\rho(\Phi_\varepsilon^{\mathrm{sq}}(0)) = 0$.

Proof. Define the "tooth" function $g : [0,1] \to [0,1]$ with

$$g(x) = \begin{cases} 2x, & \text{for } x < 1/2, \\ 2(1-x), & \text{for } x \geq 1/2, \end{cases}$$

and its composition

$$g_s(x) := \underbrace{g \circ g \circ \cdots \circ g(x)}_{s\text{-times}}.$$

Furthermore, from the definition of the ReLU function (refer to (2.15)), we can also set $g(x) = 2\rho(x) - 4\rho(x - 1/2) + 2\rho(x-1)$, which is intern, the same as defining a neural network with 1-dimensional input and output via $\Phi = ((A_1, b_1), (A_2, b_2))$ with

$$A_1 = \begin{bmatrix} \mathrm{Id}_\mathbb{R} \\ \mathrm{Id}_\mathbb{R} \\ \mathrm{Id}_\mathbb{R} \end{bmatrix}, \quad b_1 = \begin{pmatrix} 0 \\ -1/2 \\ -1 \end{pmatrix}, \quad A_2 = \begin{bmatrix} 2 & -4 & 2 \end{bmatrix}, \quad b_2 = 0.$$

Then it is evident, using Lemma (2.19), that $g_s(x) = R_\rho(\underbrace{\Phi \odot \cdots \odot \Phi}_{s\text{-times}})(x)$ for all $x \in \mathbb{R}$ represents a network with $\mathcal{O}(s)$ layers, active weights and neurons. Now, with an explicit formulation of g_m for a fixed level $m \geq 1$, one can identify the function $f(x) := x^2$ using linear interpolation on equidistant points $k/(2^m)$, where $k = 0, \ldots, 2^m - 1$, i.e. let $\Phi_m = \Phi \odot \cdots \odot \Phi$ (m-times) be a neural network, because then for all $x \in [\frac{k}{2^m}, \frac{k+1}{2^m}]$ it is true, that

$$R_\rho(\Phi_m)(x) = \left(\frac{(k+1)^2}{2^m} - \frac{k^2}{2^m} \right) \left(x - \frac{k}{2^m} \right) + \left(\frac{k}{2^m} \right)^2.$$

Why?

Then we also know that, using the Bramble-Hilbert lemma in one-dimension [BH70], the fact that $f''(x) = 2$ for all $x \in (0,1)$, and $h = 1/2^m$, that

$$\| R_\rho(\Phi_m) - f \|_{L^\infty((0,1))} \leq \max_{k=0,\dots,2^m-1} Ch^2 \| f'' \|_{L^\infty([\frac{k}{2^m}, \frac{k+1}{2^m}])} = C' 2^{-2m}.$$

Hence setting $m = \lceil \ln(1/\varepsilon) \rceil$, with $C, C' > 0$ constants, only dependent on the domain, we achieve the wanted bound from the statement for the neural network $\Phi^{\mathrm{sq}}_\varepsilon := \Phi_m$ with $\mathcal{O}(m) = \mathcal{O}(\ln(1/\varepsilon) + 1) \leq \mathcal{O}(2\ln(1/\varepsilon))$ layers, neurons and active weights. □

This statement enables us to express multiplication using a neural network. This is straight forward, when viewing the polarization identity:

$$xy = \frac{1}{2}((x+y)^2 - x^2 - y^2), \quad \text{for } x, y \in \mathbb{R}, \tag{2.1}$$

as given by the first binomial formula.

Proposition 2.24. Let $M > 0$ and $\varepsilon \in (0,1)$, then there is a ReLU neural network $\widetilde{\times}$ with 2-dimensional input and 1-dimensional output, and constants $c_1 > 0$ and $c_2 = c_2(M) > 0$, such that for $\times : \mathbb{R} \times \mathbb{R} \to \mathbb{R}; (x, y) \mapsto xy$

1. $\| R_\rho(\widetilde{\times}) - \times \|_{W^{0,\infty}((-M,M)^2)} \leq \varepsilon$,
2. if $x = 0$ or $y = 0$, then $R_\rho(\widetilde{\times})(x, y) = 0$,
3. the number of layers, active weights and neurons of $\widetilde{\times}$ is at most $c_1 \ln(1/\varepsilon) + c_2$.

Proof. Let $\delta := \varepsilon/(6M^2)$. Then we construct an approximation of the square function $f(x) = x^2$ for $x \in (0,1)$ using Proposition (2.23) for the chosen δ, i.e. we define $\Phi^{\mathrm{sq}}_\delta$, such that

$$\| R_\rho(\Phi^{\mathrm{sq}}_\delta) - f \|_{W^{0,\infty}((0,1)^1)} \leq \frac{\varepsilon}{6M^2},$$

which now of course has $\mathcal{O}(\ln(1/\delta))$ many layers, active weights and neurons. Since the statement necessarily needs to implement negative numbers as well, we show with $|x| = \rho(x) + \rho(-x)$ that it is indeed possible to expand any neural network with a positive domain to a negative domain using a neural network (by simply concatenating). Thus we construct $\widetilde{\times}$ via adding an addition layers on top of parallel square networks with scaled input, achieving

$$R_\rho(\widetilde{\times})(x, y) = 2M^2 \left(R_\rho(\Phi^{\mathrm{sq}}_\delta) \left(\frac{|x+y|}{2M} \right) - R_\rho(\Phi^{\mathrm{sq}}_\delta) \left(\frac{|x|}{2M} \right) - R_\rho(\Phi^{\mathrm{sq}}_\delta) \left(\frac{|y|}{2M} \right) \right).$$

To appreciate the third statement of our Proposition, we may now put together the necessary complexity of this network using the constants in Proposition (2.23);

1. We have 2 layers for the implementation of the absolute value, 1 for the addition in the end, and $c_0 \ln(1/\delta)$ for the square function, thus $L_{\widetilde{\times}} \leq c_0 \ln(1/\varepsilon) + c_0 \ln(6M^2) + 3$.
2. We need $4 + 2 + 2$ active weights for the first matrix to distribute each (x, y) onto the parallel networks together with 6 active weights for the superposing of the input into the $|\cdot|$-function. Another 3 are required for the last layer adding all outputs. Of course each of the different computiations for the artificial scaled square function must be in parallel, thus $\omega_{\widetilde{\times}} \leq 3c_0 \ln(1/\varepsilon) + 3c_0 \ln(6M^2) + 17$.

3. A similar consideration gives us $9 = (2+1) \cdot 3$ neurons for the first two and the last layer, thus giving us $N_{\widetilde{\times}} \leq 3c_0 \ln(1/\varepsilon) + 3c_0 \ln(6M^2) + 9$.

Finally taking the adequate maximum, and remarking, that c_0 does not depend on M, we get the wanted bound in (2).

To make the true multiplication match the form of our implemented network, using (2.1), we may rewrite for $x, y \in (-M, M)$

$$
\begin{aligned}
xy &= 4M^2 \left(\frac{x}{2M} \cdot \frac{y}{2M} \right) \\
&= 4M^2 \frac{1}{2} \left(\left(\frac{x}{2M} + \frac{y}{2M} \right)^2 - \left(\frac{x^2}{2M} \right)^2 \left(\frac{y^2}{2M} \right)^2 \right) \\
&= 2M^2 \left(\left(\frac{|x+y|}{2M} \right)^2 - \left(\frac{|x|}{2M} \right)^2 - \left(\frac{|y|}{2M} \right)^2 \right).
\end{aligned}
$$

Now with the deviation from each of the square approximation we get,

$$
\begin{aligned}
\| R_\rho(\widetilde{\times}) - \times \|_{L^\infty((0,1)^2)} &\leq 2M^2 \left(3 \cdot \sup_{\frac{|x|}{2M} : x \in (-M,M)} | R_\rho(\Phi_\delta^{\mathrm{sq}})(\widetilde{x}) - \widetilde{x}^2 | \right) \\
&\leq 6M^2 \delta = \varepsilon.
\end{aligned}
$$

We note a few things happening here for clarity; it is possible to simply use the reverse triangle equality to get all three term errors added, and then put them together. This can be done, since the $L^\infty((0,1)^2)$ norm appreciates only the maximal value of the error across all possible elements of the $(0,1)^2$ domain, but because the inner function is bijective, these are just the same. Additionally we use the, trivial by definition, statement

$$
\| g \circ f \|_{L^\infty((0,1))} \leq \| g \|_{L^\infty((0,1))}.
$$

This has shown the statement (3).

Statement (1) on the other hand, is a direct consequence of the subliminally result in Proposition (2.23). □

As a last step, before stating and proving the Yarotsky Theorem, we want to discuss the construction and complexity of a network approximating any collection of equidistantly spaced function with local support in a higher dimensional domain $(0,1)^d$ for $1 \leq d < \infty$. If we take into account, that we may also want these functions to form a partition of unity on the high dimensional domain, this will be ground to tackle the complexity of the approximation of polynomials with neural networks.

Lemma 2.25. For any $d, N \in \mathbb{N}$ there is a collection of functions

$$
\Psi = \left\{ \phi_m \,|\, m \in \{0, \dots, N\}^d \right\},
$$

with $\phi : \mathbb{R}^d \to \mathbb{R}$ for all $m = (m_1, \dots, m_d) \in \{0, \dots, N\}^d$ with the following properties:

1. $0 \leq \phi_m(x) \leq 1$ for all $\phi_m \in \Psi$ and every $x \in \mathbb{R}^d$,

2. $\sum_{\phi_m \in \Psi} \phi_m(x) = 1$ for every $x \in [0,1]^d$,

3. $\text{supp}(\phi_m) \subset B_{1/N, \|\cdot\|_{\ell^\infty}}(m/N)$ for all $\phi_m \in \Psi$,

4. there are constants $C, c \geq 1$ such that for each $\phi_m \in \Psi$ there is a neural network Φ_m with d-dimensional input and output, at most 3 layers, $C \cdot d$ active weights and neurons, such that
$$\prod_{l=1}^{d} [R_\rho(\Phi_m)]_l = \phi_m$$
and $\| [R_\rho(\Phi_m)]_l \|_{L^\infty((0,1)^d)} \leq 1$ for all $l = 1, \ldots, d$.

Proof. The idea of the proof, will be one of construction, where we will find adequate functions, which are easily transferable into ReLU neural networks. For this, define the functions
$$\psi : \mathbb{R} \to \mathbb{R}, \quad \psi(x) = \begin{cases} 1, & |x| < 1, \\ 0, & 2 < |x|, \\ 2 - |x| & 1 \leq |x| \leq 2, \end{cases}$$
and $\phi_m : \mathbb{R}^d \to \mathbb{R}$ as the prodoct of shifted and scaled versions of ψ, which inherit its local support, i.e.
$$\phi_m(x) := \prod_{l=1}^{d} \psi\left(3N\left(x_l - \frac{m_l}{N}\right)\right),$$
for $m = (m_1, \ldots, m_d) \in \{0, \ldots, N\}^d$. This function satisfies our first three parts of the Lemma directly: Take one x_i for some $i = 1, \ldots, d$. Now we get that $\psi(3N(x_i - (m_i/N))) \in [-2, 2]$, since ψ has support of $[-2, 2]$. This gives us
$$x \in \left[\frac{m}{N} - \frac{2}{3N}, \frac{m}{N} + \frac{2}{3N}\right],$$
yielding the third property, as we took i arbitrarily. Simultaneously we use the zero product property, and the fact, that the range of ψ is $[0, 1]$ to get the first property. Defining $z_i := 3Nx_i$, we can rewrite for any N the one-dimensional function
$$S(x_i) := \sum_{m_i=0}^{N} \psi\left(3N\left(x_i - \frac{m_i}{N}\right)\right) \quad \text{into the function} \quad \widetilde{S}(z_i) := \sum_{m_i=0}^{N} \psi(z_i - 3m_i),$$
that intern gives us the wanted result of $\widetilde{S}(z_i) = 1$ for any $z_i \in [0, 3N]$, because the least distance between points is 3, which means going forwards or backwards either lands one outside the support of the function, if $z_i \in [-1, 1]$, or it adds both third case values, i.e. setting $z_i \in [1, 2]$ w.l.o.g
$$\psi(z_i) + \psi(z_i - 3) = (2 - z_1) + (2 + (z_i - 3)) = 1.$$
Now we can extrapolate this result to S by a simple scaling argument and then use Fubini and the factorization of the sum over products to get
$$\sum_{m \in \{0,\ldots,N\}^d} \phi_m(x) = \sum_{m_1=0}^{N} \cdots \sum_{m_d=0}^{N} \prod_{l=1}^{d} \psi\left(3N\left(x_l - \frac{m_l}{N}\right)\right) = \prod_{l=1}^{d} \left(\sum_{m_l=0}^{N} \psi\left(3N\left(x_l - \frac{m_l}{N}\right)\right)\right) = 1.$$
Finally we are only left with proving the forth property. For this, we need to construct a neural network Φ_ψ, that realizes ψ. Let
$$A_1 := \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad b_1 := \begin{pmatrix} 2 \\ 1 \\ -1 \\ -2 \end{pmatrix}, \quad A_2 := \left[0 \,\middle|\, 1 \quad -1 \quad -1 \quad 1\right], \quad b_2 := 0,$$

define the neural network $\Phi_\psi = ((A_1, b_1), (A_2, b_2))$. With distinction by cases one can easily see, that for all $x \in \mathbb{R}$

$$\begin{aligned} R_\rho(\Phi_\psi)(x) &= A_2(\rho(A_1 x + b_1)) + b_2 \\ &= A_2\left(\begin{bmatrix}\rho(x+2) & \rho(x+1) & \rho(x-1) & \rho(x-2)\end{bmatrix}^T\right) \\ &= \rho(x+2) + \rho(x-2) - (\rho(x+1) + \rho(x-1)) \\ &= \psi(x). \end{aligned}$$

Furthermore the complexity of this network is 2 layers, 12 active weights and 6 neurons. Let $\Phi_{m,l}$ be a neural network with d-dimensional input and one-dimensional output, such that

$$R_\rho(\Phi_{m,l})(x) = 3N\left(x_l - \frac{m_l}{N}\right), \quad \text{for all } x \in \mathbb{R}^d \text{ and } m \in \{0, \ldots, N\}^d.$$

Then the complexity of this network is 1 layer, 2 active weights and $d+1$ neurons. Finally realizing that the composition of both these functions, i.e. the realization of the concatenation of networks, would yield each factor for the wanted function. If we now parallelize this concatenation, we get the forth property by definition, i.e. let

$$\Phi_m := P(\Phi_\psi \odot \Phi_{m,l} \,|\, l = 1, \ldots, d),$$

since then

$$\prod_{l=1}^{d} [R_\rho(\Phi_m)]_l(x) = \phi_m(x), \quad \text{for all } x \in \mathbb{R}^d.$$

This network now has d-dimensional input and output and the complexity, using both Lemma (2.19) and Lemma (2.21), of 3 layers, at most $(12+2) \cdot 2 \cdot d = 28d$ active weights, and at most $9d$ neurons. □

With these auxiliary results, we may finally start with the statement and proof of the Yarotsky Theorem.

Theorem 2.26 (Yarotsky Theorem). *Let $d \in \mathbb{N}, n \in \mathbb{N}_{\geq 2}, B = 1$ and $f \in \mathcal{F}_{n,d,\infty,B}$ some arbitrary function. Then there is a constant $c = c(d,n) > 0$ with the following properties:*

For any $\varepsilon \in (0,1)$, there is a neural network architecture $\mathcal{A}_\varepsilon = \mathcal{A}_\varepsilon(d,n,\varepsilon)$ with d-dimensional input and one-dimensional output such that, there is a neural network Φ_ε^f that has architecture $\mathcal{A}_\varepsilon$ and satisfies

$$\| f - R_\rho(\Phi_\varepsilon^f) \|_{W^{0,\infty}((0,1)^d)} \leq \varepsilon,$$

and

1. $L_{\mathcal{A}_\varepsilon} \leq c \cdot \ln(1/\varepsilon)$,
2. $\omega_{\mathcal{A}_\varepsilon} \leq c \cdot \varepsilon^{-d/n} \ln(1/\varepsilon)$,
3. $N_{\mathcal{A}_\varepsilon} \leq c \cdot \varepsilon^{-d/n} \ln(1/\varepsilon)$.

Proof. Given the definition of the functions $\phi_m \in \Psi$ as constructed in the proof of Lemma (2.25), we want to structure the proof of this Theorem in two sub-goals:

The formulation of an approximate function f_1 for f using the partition of unity and an adequate Taylor expansion, and the subsequent approximation of that function f_1 with a neural network Φ_{f_1} using the $\widetilde{\times}$-multiplication of Proposition (2.24).

Step 1: Taylor Expansion

For any $m \in \{0, \dots, N\}^d$, consider the $(n-1)$-Taylor polynomial for the function f at $x_0 = m/N$:

$$P_m(x) = \sum_{n:\,|n|<n} \frac{D^n f(x)}{n!}\bigg|_{x=\frac{m}{N}} \left(x - \frac{m}{N}\right)^n, \quad \text{for } x \in \mathbb{R}^d,$$

with the conventions $n! = \prod_{l=1}^d n_l!$ and $(x - m/N)^n = \prod_{l=1}^d (x_l - \frac{m_l}{N})^{n_l}$. With the partition of unity of Lemma (2.25), we can then set any function, as a linear combination of its point-values and the partition function. Especially this allows us to define

$$f = \sum_{m\in\{0,\dots,N\}^d} \phi_m f, \quad \text{and} \quad f_1 = \sum_{m\in\{0,\dots,N\}^d} \phi_m P_m.$$

Now we can exploit the properties given by Lemma (2.25), as follows; firstly use the $\| \phi_m \|_{L_\infty((0,1)^d)} = 1$ property in Line (2.3), then one can estimate the maximal number of partitions with 2^d in Line (2.4). After this we take the simple Lagrange remainder

argue this

in Line (2.5). Lastly one can simply argue with $f \in \mathcal{F}_{n,d,\infty,B=1}$ in Line (2.6).

$$\begin{aligned}
\| f - f_1 \|_{L_\infty((0,1)^d)} &\le \sum_{m\in\{0,\dots,N\}^d} \| \phi_m\,(f - P_m) \|_{L_\infty((0,1)^d)} && (2.2)\\
&\le \sum_{m:\left|x_l - \frac{m_l}{N}\right| < \frac{1}{N},\, \forall l=1,\dots,d} \| f - P_m \|_{L_\infty((0,1)^d)} && (2.3)\\
&\le 2^d \max_{m:\left|x_l - \frac{m_l}{N}\right| < \frac{1}{N},\, \forall l=1,\dots,d} \| f - P_m \|_{L_\infty((0,1)^d)} && (2.4)\\
&\le \frac{2^d d^n}{n!} \left(\frac{1}{N}\right)^n \max_{n:\,|n|=n} \| D^n f \|_{L_\infty((0,1)^d)} && (2.5)\\
&\le \frac{2^d d^n}{n!} \left(\frac{1}{N}\right)^n && (2.6)
\end{aligned}$$

Now taking

$$N = \left\lceil \left(\frac{n!}{2^d d^n} \cdot \frac{\varepsilon}{2}\right)^{-\frac{1}{n}} \right\rceil$$

gives the approximation bound of

$$\| f - f_1 \|_{L_\infty((0,1)^d)} \le \frac{\varepsilon}{2}. \tag{2.7}$$

Step 2: Approximation by a Neural Network

The polynomial function f_1 may be further expressed in a different form, which allows us to seperately compare it to a potential network approximation using the definition of ϕ_m, i.e. for $x \in \mathbb{R}^d$

$$f_1(x) = \sum_{m\in\{0,\dots,N\}^d} \sum_{n:\,|n|<n} a_{m,n} \underbrace{\left(\phi_m(x)\left(x - \frac{m}{N}\right)^n\right)}_{=:f_{m,n}(x)}. \tag{2.8}$$

This expansion now has at most $d^n(N+1)^d$ terms containing $f_{m,n}$, since

$$\left|\left\{n \in \{0,\dots,n-1\}^d \,:\, |n| < n\right\}\right| \leq \left|\left\{n \in \{0,\dots,n-1\}^d\right\}\right| = d^n,$$

and obviously $|\{m = \{0,\dots,N\}^d\}| = (N+1)^d$. We note, that we take $M = d+n$ for the definition of the network multplication, i.e. $\widetilde{\times} : (-M,M)^2 \to \mathbb{R}$. This is done, because we have as an obvious consequence from the fact, that if $x,y \in (-1,1)$, that

$$|\, R_\rho(\widetilde{\times})(x,y) - xy \,| \leq \delta \qquad \Longrightarrow \qquad |\, R_\rho(\widetilde{\times})(x,y) \,| \leq |y| + \delta,$$

for some fixed but not yet specified $\delta \in (0,1)$. Recursively concatenating $\widetilde{\times}$ for $(d+n-1)$-times can then propagate the error to at most $M = (d+n)$ upon the last realization of $\widetilde{\times}$.

As aforementioned, we can now compare each function $f_{m,n}$ pointwise to a realization of a network approximating at most $n-1$ of the multiplications needed for $\prod_{l=1}^d (x_l - (m_l/N))^{n_l}$, as we know that $\sum_{l=1}^d n_l < n$, d additional multiplications for the realization of the function ϕ_m and one multiplication for putting it together. Hence we define a neural network $\Phi_{f_1}^{m,n}$ with d-dimensional input and one dimensional output via

$$\begin{aligned} R_\rho(\Phi_{f_1}^{m,n})(x) = R_\rho(\widetilde{\times})\Big([R_\rho(\Phi_m)]_1(x_1), R_\rho(\widetilde{\times})\Big([R_\rho(\Phi_m)]_2(x_2), \dots \\ R_\rho(\widetilde{\times})\Big([R_\rho(\Phi_m)]_d(x_d), R_\rho(\widetilde{\times})\Big(x_1 - \frac{m_1}{N}, \dots \\ R_\rho(\widetilde{\times})\Big(x_{d-1} - \frac{m_{d-1}}{N}, R_\rho(\widetilde{\times})\Big(x_d - \frac{m_d}{N}, \dots \\ R_\rho(\widetilde{\times})\Big(x_d - \frac{m_d}{N}, x_d - \frac{m_d}{N} \Big)\Big)\Big)\Big)\Big)\Big)\Big). \end{aligned}$$

for $x = (x_1,\dots,x_d) \in \mathbb{R}^d$ where we used Φ_m as in the proof Lemma (2.25). This network contains at most $(d+n)$ concatenations of $\widetilde{\times}$, and thus by Lemma (2.25) and its constant $C > 0$, Lemma (2.19), and Proposition (2.24) and its constants $c_0, c_1(d,n) > 0$,

$$L_{\Phi_{f_1}^{m,n}} \leq 3c_0(d+n)\left(\ln\left(\frac{1}{\delta}\right) + (d+n)c_1\right), \quad \omega_{\Phi_{f_1}^{m,n}} \leq C2(d+n)d\left(\ln\left(\frac{1}{\delta}\right) + 2(d+n)C\right). \tag{2.9}$$

for a given precision level $\delta \in (0,1)$, where the number of neurons are of the same order as the number of active weights.

The error comparison of $R_\rho(\Phi_{f_1}^{m,n})$ and $f_{m,n}$ can then be done by counting the maximal depth of usages of the network $\widetilde{\times}$, which is $(d+n)$ as shown above, and the estimate of

$$\left|\, \left(x - \frac{m}{N}\right) \,\right| \leq 1, \quad |\, \psi(x) \,| \leq 1,$$

for all $x \in (0,1)^d$. Then directly

$$\|\, R_\rho(\Phi_{f_1}^{m,n}) - f_{m,n} \,\|_{L_\infty((0,1)^d)} \leq (d+n)\delta.$$

Now since, as mentioned below Equation (2.8), f_1 has at most $d^n(N+1)^d$ terms, but since any $x \in \mathbb{R}^d$ can be only in the support of at most 2^d functions ϕ_m by the second statement in Proposition (2.24), the sum going over all m can be reduced to only those, which are non-zero, which we employ in Line (2.11). This brings $(N+1)^d \to 2^d$, supporting the equation in Line (2.12). Setting

$$\delta = \frac{\varepsilon}{2 \cdot 2^d d^n (d+n)}$$

allows us to finally assert, using the bound of Equation (2.9) for Line (2.13),

$$
\begin{aligned}
\| R_\rho(\Phi_{f_1}) - f_1 \|_{L_\infty((0,1)^d)} &\leq \sum_{m \in \{0,\dots,N\}^d} \sum_{n\colon |n|<n} |a_{m,n}| \, \| R_\rho(\Phi_{f_1}^{m,n}) - f_{m,n} \|_{L_\infty((0,1)^d)} && (2.10)\\
&= \sum_{m\colon x \in \operatorname{supp}(\phi_m)} \sum_{n\colon |n|<n} |a_{m,n}| \, \| R_\rho(\Phi_{f_1}^{m,n}) - f_{m,n} \|_{L_\infty((0,1)^d)} && (2.11)\\
&\leq 2^d d^n \max_{m\colon x \in \operatorname{supp}(\phi_m)} \max_{n\colon |n|<n} \| R_\rho(\Phi_{f_1}^{m,n}) - f_{m,n} \|_{L_\infty((0,1)^d)} && (2.12)\\
&\leq 2^d d^n (d+n)\delta && (2.13)\\
&= \frac{\varepsilon}{2}. && (2.14)
\end{aligned}
$$

Our knowledge of δ together with at most $d^n(N+1)^d = \mathcal{O}(\varepsilon^{-d/n})$ parallel networks with one additive layer and the fact that generally we have $c' \ln(1/\delta + c'') \leq c''' \ln(1/\delta)$ for $c', c'' > 0$ and $c''' \geq c' \ln(1/c'')$, gets us the final complexity of Φ_{f_1} from the starting point of Equation (2.9), i.e.

$$L_{\Phi_{f_1}} \leq c \ln(1/\varepsilon), \quad \omega_{\Phi_{f_1}} \leq c\varepsilon^{-d/n} \ln(1/\varepsilon), \quad N_{\Phi_{f_1}} \leq c\varepsilon^{-d/n} \ln(1/\varepsilon).$$

with some constant $c = c(d,n) > 0$.

Using the triangle inequality, we can show that this network indeed satisfies the wanted precision bound with both errors stemming from Equation () and Equation () respectively:

$$\| f - R_\rho(\Phi_{f_1}) \|_{L_\infty((0,1)^d)} \leq \| f - f_1 \|_{L_\infty((0,1)^d)} + \| f_1 - R_\rho(\Phi_{f_1}) \|_{L_\infty((0,1)^d)} \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

We may even expand Yarotsky to cover a more general case, where $f \in \mathcal{F}_{n,d,p,B}$ for arbitrarily $1 \leq p \leq \infty$ and $B > 0$ and the norm, to which we assert the precision, an arbitray Sobolev norm. This extension is presented by Gühring et al. [GKP19].

Theorem 2.27 (Gühring Theorem). *Let $d \in \mathbb{N}, n \in \mathbb{N}_{\geq 2}, B > 0$ and $0 \leq s \leq 1$ and $f \in \mathcal{F}_{n,d,p,B}$ some arbitrary function. Then there is a constant $c = c(d,n,p,B,s) > 0$ with the following properties:*

For any $\varepsilon \in (0,1)$, there is a neural network architecture $\mathcal{A}_\varepsilon = \mathcal{A}_\varepsilon(d,n,p,B,s,\varepsilon)$ with d-dimensional input and one-dimensional output such that, there is a neural network Φ_ε^f that has architecture $\mathcal{A}_\varepsilon$ and satisfies

$$\| f - R_\rho(\Phi_\varepsilon^f) \|_{W^{s,p}((0,1)^d)} \leq \varepsilon,$$

and

1. $L_{\mathcal{A}_\varepsilon} \leq c \cdot \ln(1/\varepsilon)$,
2. $\omega_{\mathcal{A}_\varepsilon} \leq c \cdot \varepsilon^{-d/(n-s)} \ln(1/\varepsilon)$,
3. $N_{\mathcal{A}_\varepsilon} \leq c \cdot \varepsilon^{-d/(n-s)} \ln(1/\varepsilon)$.

Proof Sketch. Considering that all of the tools used for this proof will neither be of any use to the main results in this paper nor to the general structure, we will only sketch the general steps necessary to achieve it for the interested reader:

1. The first step is to expand the Proposition (2.23) to the $W^{1,\infty}(0,1)$-norm. This is done by simple calculation, since an explicit construction is being made.

2. This can be applied to Proposition (2.24) in the same way. Additionally we need to bound $|\, R_\rho(\widetilde{\times}) \,|_{W^{1,\infty}}((0,1)^2)$, since this is required in for the bound on the collective error under the Taylor expanded function in the end.

3. The most import additional step is to bound the switch of norms from $W^{1,p}((0,1)^d)$ to $W^{s,p}((0,1)^d)$ of a function being approximated by the partition of unity, that we discussed in Lemma (2.25). This gives

$$\| \, f - f_N \, \|_{W^{s,p}((0,1)^d)} \leq C \left(\frac{1}{N} \right)^{n-s} \| \, f \, \|_{W^{n,p}((0,1)^d)},$$

   for a partition of unity of size N. It is worth noting, that this factor in front is responsible for the different approximation in regard to weights and neurons, since it influences the necessary choice of δ.

4. The last step, which differs greatly from the proof of Yarotsky Theorem (2.26), is the approximation of a sum of localized polynomials with a neural network. Here the challenge lies again in the control of the derivatives.

□

Chapter 3

Formulation of the General Problem

Describe basic already existing techniques via literature

Introduce Proper Orthogonal Decomposition

Introduce Autoencoders

Introduce Deep Orthogonal Decomposition

3.1 Time-dependent parametric PDE

Brief general description of time dependent parametric PDEs regarding existing literature.

Definitely add traditional convergence and computational cost.

Let $\Theta \subset \mathbb{R}^p$ and $\Theta' \subset \mathbb{R}^q$ denote the geometric and physical parameter spaces respectively. Further we let $\Omega \subset \mathbb{R}^d$ be some Lipschitz domain and $\Gamma = [0, T]$ for some $T > 0$ be the trial space of a parametric time-dependent PDE, i.e. $u^{\mu,\nu}(x,t) := u(x,t;\mu,\nu)$ is the solution of the PDE, if

$$\begin{cases} \partial_t u^{\mu,\nu}(x,t) + \mathcal{L}(\mu,\nu)u^{\mu,\nu}(x,t) + \mathcal{N}(u^{\mu,\nu}(x,t),\mu,\nu) = f(x,t;\mu,\nu), & (x,t) \in \Omega \times \Gamma, \\ \mathcal{B}(\mu,\nu)u^{\mu,\nu}(x,t) = g_N(x,t;\mu,\nu) & (x,t) \in \partial\Omega \times \Gamma, \\ u^{\mu,\nu}(x,t) = u_0^{\mu,\nu}(x) & (x,t) \in \Omega \times \{0\}. \end{cases}$$

Assume we half-discretize the problem in the spacial domain Ω with N_h degrees of freedom. Note that we refer to this discrete grid space on Ω as Ω^h and its boundary grid points as $\partial\Omega^h$. Then the resulting full order model can be seen as the solution to

$$\begin{cases} M(\mu,\nu)\partial_t u_h(t;\mu,\nu) + A(\mu,\nu)u_h(t;\mu,\nu) + N(u_h(\mu,\nu),\mu,\nu) = f(t;\mu,\nu) & (x,t) \in \Omega^h \times \Gamma, \\ u_h(0;\mu,\nu) = u_0(\mu,\nu) & (x,t) \in \Omega^h \times \{0\}, \end{cases}$$

where we assume $M : \Theta \times \Theta' \to \mathbb{R}^{N_h \times N_h}$ to send the parameter set to a mass matrix, $A : \Theta \times \Theta' \to \mathbb{R}^{N_h \times N_h}$ to a symmetric positive definite stiffness matrix, $N : \mathbb{R}^{N_h} \to \mathbb{R}^{N_h}$ to be some non-linear term and $f : \Gamma \times \mathbb{R}^{N_h} \times \Theta \times \Theta' \to \mathbb{R}^{N_h}$ to be the corresponding source term of the equation. Additionally we assume to have initial conditions via the function $u_0 : \Theta \times \Theta' \to \Omega^{N_h}$.

Assumption 1. Let $\mathcal{S} := \{u(\mu,\nu,t) \,|\, (\mu,\nu,t) \in \Theta \times \Theta' \times \mathcal{T}\}$ denote the total solution manifold and $\mathcal{S}_{\mu,t} := \{u(\mu,\nu,t) \,|\, \nu \in \Theta'\}$ denote the solution submanifold for a fixed set $(\mu,t) \in \Theta \times \mathcal{T}$, then we assume that

1. the linear KnW of $\mathcal{S}$ decays slowly, i.e.

$$d_N(\mathcal{S}) \leq CN^{-\alpha},$$

where $0 < \alpha \leq 1$ and $C > 0$ some constant.

2. the linear KnW of $\mathcal{S}_{\mu,t}$ decays quickly for all $\mu \in \Theta$ and $t \in \mathcal{T}$, i.e.

$$\sup_{(\mu,t)\in\Theta\times\mathcal{T}} d_N(\mathcal{S}_{\mu,t}) \leq C'N^{-\beta},$$

where $\beta > 1$ and $C' > 0$ some constant.

3.2 Deep-learning based Reduced Order Models

Introduce POD-DL-ROM

A general Deep learning based reduced order model (DL-ROM) firstly appeared in architecture and training in the paper [FDM20]; to properly approximate the potentially non-linear solution manifold of the high fidelity FOM, we employ an Autoencoder (AE), and on the reduced manifold we employ some other Deep Learning architecture, like the proposed Deep Feed-Forward Neural Network (DFNN), to learn the latent dynamics of the parametric PDE. This concludes in the following approximation of the FOM solution u_h

$$\hat{u}_h(t;\mu,\nu) = \Psi_h(\Phi_n(t;\mu,\nu,\theta_{DF}),\theta_D) \in \mathbb{R}^{N_h}, \tag{3.1}$$

where we set $\Psi_h = f_h^D : \mathbb{R}^n \times \Theta_D \to \mathbb{R}^{N_h}$ for the decoder (and $f_n^E : \mathbb{R}^{N_h} \times \Theta_E \to \mathbb{R}^n$ the corresponding encoder) of a convolutional AE. Then f_n^E denotes the projection onto the reduced trial solution manifold and we assume $\Phi_n = \Phi_n^{DF} : \Gamma \times \Theta \times \Theta' \times \Theta_{DF} \to \mathbb{R}^n$ to be the DFNN mapping the dynamics of the parametric PDE onto the reduced trial solution manifold.

The subsequent training is then performed via a loss function of a convex combination of both, the projection error using the encoder and the dynamics error considering both the decoder and the feed forward network, i.e. for $\theta = (\theta_D, \theta_E, \theta_{DF})$ define the optimal weights via

$$\theta^* = \arg\min_{\theta} \frac{1}{N_s} \sum_{i,j=1}^{N_{train}} \sum_{k=1}^{N_t} \frac{\omega_h}{2} \parallel u_h(t^k;\mu_i,\nu_j) - f_h^D(\Phi_n^{DF}(t^k;\mu_i,\nu_j,\theta_{DF}),\theta_D),\theta_D) \parallel^2 \tag{3.2}$$

$$+ \frac{1-\omega_h}{2} \parallel f_n^E(u_h(t^k;\mu_i,\nu_j),\theta_E) - \Phi_n^{DF}(t^k;\mu_i,\nu_j,\theta_{DF}) \parallel^2. \tag{3.3}$$

3.2.1 Deep Orthogonal Decomposition based Reduction

Introduce both DOD-DL-ROM and DOD+DNN

For comparison, one can see, that the POD-DL-ROM works significantly more efficient than the DL-ROM by pre-reducing the dimension N_h to some $N < N_h$ in a linear POD-based fashion. This has the disadvantage of potentially losing fidelity, if the Kolmogorov N-width is rather large, which is the case for non-linear PDEs, relying on geometric circumstances. For this, we might instead of using snapshot-based rPOD for the matrix S (optionally the correlation matrix) rather directly employ a Deep Orthogonal Decomposition to find the reduction matrix in dependency of $\mu \in \Theta$ and $t \in \Gamma$.

Thus we have a common denominator for the DOD-DL-ROM approach. Defining $V : \Gamma \times \Theta \times \Theta_{DOD} \to \mathbb{R}^{N_h \times N}; V(t;\mu,\theta_{DOD}) \mapsto \mathbb{A}\widetilde{V}_{N_A}(t;\mu,\theta_{DOD})$, where $\mathbb{A} \in \mathbb{R}^{N_h \times N_A}$ is the pre-reduction

POD for $N < N_A < N_h$ employed to deal with exceptionally large inherent dimensions N_h and $\widetilde{V}_{N_A}(t;\mu,\theta_{DOD}) \in \mathbb{R}^{N_A \times N}$ is called *inner module* of the DOD, yields the foundation of the reduction consideration. From hereon, there are a couple of different approaches, which lead to similar ideas; one is able to, after learning the DOD-map, either suppose, the remaining reduced solution manifold is "linear" enough to work directly with a coefficient finder network (Option A) or a further non-linear reduction, potentially by an Autoencoder, is needed (Option B). As an additional, maybe more ambiguous regarding error estimates, we want to introduce a CoLoRA-based ROM, which first tries to use the DOD-NN reductor for the stationary equivalent of the problem (if existent) and then employs a CoLoRA architecture to reduce the dimension up to the dynamics of the system (Option C).

For Options A and B: To use the same loss function as in the (POD-)DL-ROM, we do need to guarantee a previous construction of the inner module of the DOD. This first step is done with a separate training with a time dependent analogue to the DOD, i.e.

$$\theta^*_{DOD} = \arg\min_{\theta_{DOD}} \frac{1}{N_s} \sum_{i,j=1}^{N_{train}} \sum_{k=1}^{N_t} \| u_h(t^k;\mu_i,\nu_j) - V(t^k;\mu_i,\theta_{DOD}) V^T(t^k;\mu_i,\theta_{DOD}) \mathbb{G} u_h(t^k;\mu_i,\nu_j) \|^2. \tag{3.4}$$

3.2.2 DOD+DNN

Let $\Phi_n^{DLNN} : \Gamma \times \Theta \times \Theta' \times \Theta_{DLNN} \to \mathbb{R}^N$ be some Deep Feed-Forward Neural Network, then the overall workflow can be summed up by

$$\hat{u}_h(t;\mu,\nu,\theta_{DOD},\theta_{DFNN}) = \mathbb{A}\widetilde{V}_{N_A}(t;\mu,\theta_{DOD}) \cdot \Phi_n^{DFNN}(t;\mu,\nu,\theta_{DFNN}). \tag{3.5}$$

This is a rather obvious extension of [Fra+24], but works surprisingly well. It will be the basic benchmark to overcome for the following algorithms. As for training a general comparison between the projection of a true solution and the prediction of the Feed Forward Network works well, i.e. for $\theta = \theta_{DFNN}$ let the optimal weights be given by

$$\theta^* = \arg\min_{\theta} \frac{1}{N_s} \sum_{i,j=1}^{N_{train}} \sum_{k=1}^{N_t} \| V^T(t^k;\mu_i,\theta^*_{DOD}) u_h(t^k;\mu_i,\nu_j) - \Phi_n^{DFNN}(t^k;\mu_i,\nu_j,\theta_{DFNN}) \|. \tag{3.6}$$

3.2.3 DOD-DL-ROM

Let f_N^D and f_n^E be the decoder and encoder of an AE respectively as above. Each of these non-linear DOD-DL-ROMs is subject to further dimensional reduction and thus is not comparable directly to the linear method above. Nonetheless in terms of reduction in dimensionality both methods can be comparable, if the latent dynamics dimension n is the same in both cases. The general workflow is given by

$$\hat{u}_h(t;\mu,\nu,\theta_{DOD},\theta_D,\theta_{DFNN}) = \mathbb{A}\widetilde{V}_{N_A}(t;\mu,\theta_{DOD}) \cdot f_N^D(\Phi_n^{DFNN}(t;\mu,\nu,\theta_{DFNN}),\theta_D). \tag{3.7}$$

Since for this, we employ an AE, we optimize for two loss functions, since the employed decoder is tied strictly to its respective encoder being able to correctly project onto the low-dimensional subspace of the latent dynamics of the PDE. For $\theta = (\theta_D, \theta_E, \theta_{DFNN})$ define the optimal weights via

$$\theta^* = \arg\min_{\theta} \frac{1}{N_s} \sum_{i,j=1}^{N_{train}} \sum_{k=1}^{N_t}$$
$$\frac{\omega_h}{2} \parallel u_h(t^k;\mu_i,\nu_j) - V(t^k;\mu_i,\theta^*_{DOD}) \cdot f_h^D(\Phi_n^{DFNN}(t^k;\mu_i,\nu_j,\theta_{DFNN}),\theta_D),\theta_D) \parallel^2$$
$$+ \frac{1-\omega_h}{2} \parallel V^T(t^k;\mu_i,\theta^*_{DOD}) \cdot f_n^E(u_h(t^k;\mu_i,\nu_j),\theta_E) - \Phi_n^{DFNN}(t^k;\mu_i,\nu_j,\theta_{DFNN}) \parallel^2 .$$

3.2.4 CoLoRA-DL-ROM

Introduce CoLoRA DL-ROM

Here we aim at minimizing the dimension through two major reductions; the first being the reduction of the stationary DOD to the latent dimension of the μ-invariant solution manifold of the stationary equivalent to the problem (regarding the Dirichlet boundary conditions as the initial conditions on some a priori defined boundary), and the second concerning the actual reduction with CoLoRA-layered architecture to the latent dynamic dimension $n << N_h$. The general workflow is thus given by

$$\hat{u}_h(t;\mu,\nu,\theta^*_{sDOD},\theta_C) = \mathbb{A} \cdot \mathcal{C}_L(\ldots(\mathcal{C}_2(\mathcal{C}_1(V_0(\mu,\theta^*_{sDOD}) \cdot \phi_0(\mu,\nu,\theta^*_{sDOD})),\theta_{C_1}),\theta_{C_2}),\theta_{C_L}), \quad (3.8)$$

where $\mathcal{C}_i : \mathbb{R}^{N_A} \times \Theta_C \to \mathbb{R}^{N_A}; \mathcal{C}_i(x,\theta_{C_i}) = W_i x + A_i \operatorname{diag}(\alpha_i(\nu,t)) B_i x + b_i$ for all $1 \leq i \leq L$. Remark, that we will still employ a pre-reduction using the rPOD matrix $\mathbb{A}$ stemming from the same snapshot matrix S as in both Options before.

Chapter 4

Approximation Theory for Reduced Order Models

Include proof and discussion for general DL-ROMs

The main goal of all error estimates can be generally hindered by two rather obvious circumstances: bad sampling and a completely rigid parameter-to-solution map. Thus two main assumptions will be assumed throughout the following section.

Assumption 2. Let $p, q > 0$, assume that $\Theta \subset \mathbb{R}^p$ and $\Theta' \subset \mathbb{R}^q$ are compact and denote $\mathcal{T} = [0, T]$ for some $T > 0$. We assume that all N_s data snapshots are sampled uniformly and iid in the parameter spaces Θ, Θ', while a uniform grid is employed for the time variable, $t \in \{\Delta t, 2\Delta t, \ldots, N_t \Delta t\}$, where $N_t \in \mathbb{N}_{\geq 2}$, N_{s_1} and N_{s_2} are the respective number of random samples in the parameter spaces and $N_s = N_{s_1} N_{s_2} \in \mathbb{N}$ the total sample size, and $\Delta t = T/N_t$.

We remark, that the equidistant spacing of the time samples is a matter of convenience, and could be loosened under some constraints.

Assumption 3. Let $\mathcal{G} : \Theta \times \Theta' \to \mathbb{R}^{N_h}$ be the parameter-to-solution map, mapping $(\mu, \nu, t) \mapsto u_h(\mu, \nu, t)$. Here we assume that

1. $m = \underset{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}}{\text{ess sup}} \| u_h(\mu, \nu, t) \| > 0$, $M = \underset{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}}{\text{ess inf}} \| u_h(\mu, \nu, t) \| < \infty$,

2. $\mathcal{G}$ is Lipschitz-continuous with the constant $L > 0$.

Assumption 4. We assume that for s, s' sufficiently large, there are two maps $\psi_* : \mathbb{R}^n \to \mathbb{R}^N$ and $\psi'_* : \mathbb{R}^N \to \mathbb{R}^n$ which attain the perfect nonlinear Kolmogorov n-width of the reduced solution manifold of the problem, as given in (2.10) for any $n \leq 2(p + q) + 3$, i.e. for $\mathcal{S}_N := \{q(\mu, \nu, t) = V^T u(\mu, \nu, t) \,|\, (\mu, \nu, t) \in \Theta \times \Theta' \times \mathcal{T}\}$, where $V \in \mathbb{R}^{N_h \times N}$ is the POD matrix,

$$\delta_n(\mathcal{S}_N) = 0.$$

Proposition 4.1. Let $f \in L^2(\Theta \times \Theta' \times \mathcal{T})$. Under Assumption (2), it is true, that

$$\mathbb{E}\left[\left| \int_{\Theta\times\Theta'\times\mathcal{T}} f(\mu, \nu, t) d(\mu, \nu, t) - \frac{|\Theta \times \Theta' \times \mathcal{T}|}{N_{data}} \sum_{i=1}^{N_{s_1}} \sum_{j=1}^{N_{s_2}} \sum_{k=1}^{N_t} f(\mu_i, \nu_j, t_k) \right|\right] \leq \mathcal{O}(N_s^{-1/2} + N_t^{-1}).$$

4.1 Relative Error of POD+DNNs

Discuss relative errors for the plain POD+DNN

Theorem 4.2. *Let $\mathcal{G} : (\mu, \nu, t) \mapsto u_h(\mu, \nu, t)$ for all $\mu \in \Theta, \nu \in \Theta'$ and $t \in \mathcal{T}$ be the parameter-to-solution map. Here we consider a general POD+DNN structure, thus neglecting the potentially different nature of Θ and Θ', i.e. $\mathcal{G}(\mu, \nu, t) \approx V\hat{q}$, where $\hat{q} : \mathbb{R}^{p+q+1} \to \mathbb{R}^N$ is a neural network trained using data $(\mu_i, \nu_i, t_i)_{i=1}^{N_{data}}$ and V is the POD matrix, as described in (2.6). Then under the Assumptions (2) and (3), we have*

$$\mathcal{E}_R \leq \mathcal{E}_S + \mathcal{E}_{POD} + \mathcal{E}_{NN}, \tag{4.1}$$

where

1. *The sampling error*
$$\mathcal{E}_S = \mathcal{E}_S\left(\mathcal{G}, (\mu_i, \nu_i, \tau_i)_{i=1}^{N_{data}}, N\right)$$
satisfies $\mathcal{E}_S \to 0$ as $N_s, N_t \to \infty$, with expectation
$$\mathbb{E}[\mathcal{E}_S] = \mathcal{O}(N_s^{-1/4} + N_t^{-1}).$$

2. *The POD projection error*
$$\mathcal{E}_{POD} = \mathcal{E}_{POD}\left(\mathcal{G}, (\mu_i, \nu_i, \tau_i)_{i=1}^{N_{data}}, N\right)$$
satisfies $\mathcal{E}_{POD} \to \mathcal{E}_{POD,\infty}$ almost surely as $N_s, N_t \to \infty$, where
$$\mathcal{E}_{POD,\infty} = \mathcal{E}_{POD,\infty}(\mathcal{G}, N)$$
is independent of the data snapshots and proportional to the linear KnW of the manifold $\mathcal{S} = \{u(\mu, \nu, t) \,|\, (\mu, \nu, t) \in \Theta' \times \mathcal{T}\}$, i.e. $d_N(\mathcal{S})$.

3. *The neural network approximation error*
$$\mathcal{E}_{NN} = \mathcal{E}_{NN}(\mathcal{G}, N, \hat{q})$$
is arbitrarily small depending on the complexity and structure of the neural network $\hat{q}$.

Proof. With the knowledge of $\|\cdot\|_{L^2_\omega}$ being a norm, we know it must be subject to the triangle inequality, thus

$$\begin{aligned} \mathcal{E}_R &= \left(\int_{\Theta\times\Theta'\times\mathcal{T}} \frac{\| u(\mu,\nu,t) - V\hat{q}(\mu,\nu,t) \|}{\| u(\mu,\nu,t) \|} d(\mu,\nu,t)\right)^{1/2} && (4.2)\\ &\leq \| u(\mu,\nu,t) - VV^T u(\mu,\nu,t) \|_{L^2_\omega} + \| VV^T u(\mu,\nu,t) - V\hat{q}(\mu,\nu,t) \|_{L^2_\omega}. && (4.3)\end{aligned}$$

Let $q(\mu, \nu, t) := V^T u(\mu, \nu, t)$ be a reduced solution according to the POD basis. Then the natural definition of the neural network approximation error is

$$\mathcal{E}_{NN} = \left(\int_{\Theta\times\Theta'\times\mathcal{T}} \frac{\| Vq(\mu,\nu,t) - V\hat{q}(\mu,\nu,t) \|^2}{\| u(\mu,\nu,t) \|^2} d(\mu,\nu,t)\right)^{1/2}. \tag{4.4}$$

This marks the second part in (4.3). This is only dependent on the ability of the neural network to identifying the best choice on a given linear sub-manifold of the solution manifold. The first part is concerned with the error generated from the best *possible* approximation of such a linear sub-manifold.

For this, we can bound the norm $\| \cdot \|_{L^2_\omega} \leq m^{-1} \| \cdot \|$, where $m > 0$ is the constant stemming from Assumption (3). Further we remind ourselves of the definition of the discrete correlation matrix $K = |\Theta \times \Theta \times \mathcal{T}| N_{data}^{-1} U U^T$ and its eigenvalues σ_k in (2.6). Using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ for all $a, b \geq 0$, we get

$$
\begin{aligned}
&\| u(\mu,\nu,t) - VV^T u(\mu,\nu,t) \|_{L^2_\omega} \\
&\leq m^{-1} \left(\int_{\Theta\times\Theta'\times\mathcal{T}} \| u(\mu,\nu,t) - VV^T u(\mu,\nu,t) \|^2 d(\mu,\nu,t) \right)^{1/2} \\
&\leq m^{-1} \left(\int_{\Theta\times\Theta'\times\mathcal{T}} \| u(\mu,\nu,t) - VV^T u(\mu,\nu,t) \|^2 d(\mu,\nu,t) - \sum_{k>N} \sigma_k^2 + \sum_{k>N} \sigma_k^2 \right)^{1/2} \\
&\leq m^{-1} \left(\left| \int_{\Theta\times\Theta'\times\mathcal{T}} \| u(\mu,\nu,t) - VV^T u(\mu,\nu,t) \|^2 d(\mu,\nu,t) - \sum_{k>N} \sigma_k^2 \right| + \sum_{k>N} \sigma_k^2 \right)^{1/2} \\
&\leq \underbrace{m^{-1} \left| \int_{\Theta\times\Theta'\times\mathcal{T}} \| u(\mu,\nu,t) - VV^T u(\mu,\nu,t) \|^2 d(\mu,\nu,t) - \sum_{k>N} \sigma_k^2 \right|^{1/2}}_{=:\mathcal{E}_S} + \underbrace{m^{-1} \sqrt{\sum_{k>N} \sigma_k^2}}_{=:\mathcal{E}_{POD}}.
\end{aligned}
$$

This concludes the estimate up to the facts about the convergence of the errors. Recalling (2.8), we can rewrite the sampling error as follows

$$
\begin{aligned}
\mathcal{E}_S = m^{-1} \Bigg| \int_{\Theta\times\Theta'\times\mathcal{T}} &\| u(\mu,\nu,t) - VV^T u(\mu,\nu,t) \|^2 \, d(\mu,\nu,t) - \\
&\frac{|\Theta\times\Theta'\times\mathcal{T}|}{N_{data}} \sum_{j=1}^{N_{data}} \| u_j - VV^T u_j \|^2 \Bigg|^{1/2}
\end{aligned}
$$

Furthermore, we define using the compactness postulated in Assumption (2) and the boundedness in Assumption (3), the L^2-error map via

$$
f(\mu,\nu,t) := \| u(\mu,\nu,t) - VV^T u(\mu,\nu,t) \|^2 \leq M^2 \| I - VV^T \|^2 < \infty,
$$

where we justify neglecting the dependence on u in the mapping, since $\mathcal{G}(\mu,\nu,t)$ is assumed to be known and can be projected onto the N_h-dimensional solution manifold. Thus with $f \in L^2(\Theta\times\Theta'\times\mathcal{T})$, we know that by Proposition (4.1), $\mathbb{E}[\mathcal{E}_S] \leq \mathcal{O}(N_s^{-1/4} + N_t^{-1/2})$.

By the strong law of large numbers, we can thus conclude, that $\mathcal{E}_S \to 0$ almost surely, as $N_t, N_s \to \infty$, and further by (2.7), we have that $\mathcal{E}_{POD} \to \mathcal{E}_{POD,\infty}$ as $N_t, N_s \to \infty$, where

$$
\mathcal{E}_{POD,\infty} = m^{-1} \sqrt{\sum_{k>N} \sigma_{k,\infty}^2}. \tag{4.5}
$$

□

Theorem 4.3. *Under the assumptions of Theorem (4.2), we have that*

$$\mathcal{E}_R \geq \frac{m}{M}\mathcal{E}_{DOD,\infty},$$

where $\mathcal{E}_{DOD,\infty} := m^{-1}\sqrt{\sum_{k>N}\sigma_{k,\infty}^2}$.

Proof.

Proof this

□

4.2 Relative Error of DOD+DNNs

Extrapolate to DOD+DNN

Lemma 4.4 (Existence of the DOD-Module)**.**

Rework the proof and statement according to the setting

Let $\Theta \subset \mathbb{R}^p$ and $\Theta' \subset \mathbb{R}^q$ be two compact sets. Let

$$\mathcal{F} : \Theta \times \Theta' \to \mathbb{R}^{N_h};\ (\boldsymbol{\mu}, \boldsymbol{\nu}) \mapsto \boldsymbol{u}_{\boldsymbol{\mu},\boldsymbol{\nu}}$$

be continuous. For each $\boldsymbol{\mu} \in \Theta$, let $\mathcal{S}_{\boldsymbol{\mu}} := \{\boldsymbol{u}_{\boldsymbol{\mu},\boldsymbol{\nu}}\}_{\boldsymbol{\nu}\in\Theta'} \subset \mathcal{F}(\Theta \times \Theta')$ be the $\boldsymbol{\mu}$-submanifold in the image of $\mathcal{F}$. Let $\mathbb{G}$ be the Gram matrix associated with an inner product in $\mathbb{R}^{N_h}$, and let $\|\cdot\|$ be its corresponding norm. Then, for every $\varepsilon > 0$ there is a ReLU matrix-valued deep neural network $\boldsymbol{V} : \mathbb{R}^p \to \mathbb{R}^{N_h \times n}$ such that

$$\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{\nu}}\| \boldsymbol{u}_{\boldsymbol{\mu},\boldsymbol{\nu}} - \boldsymbol{V}_{\boldsymbol{\mu}}\boldsymbol{V}_{\boldsymbol{\mu}}^T\mathbb{G}\boldsymbol{u}_{\boldsymbol{\mu},\boldsymbol{\nu}} \| < \varepsilon + \mathbb{E}_{\boldsymbol{\mu}}[d_n(\mathcal{S}_{\boldsymbol{\mu}})]$$

where $\boldsymbol{V}_{\boldsymbol{\mu}} := \boldsymbol{V}(\boldsymbol{\mu})$.

Proof. Let $n \in \mathbb{N}$ be fixed. Assuming we have proven, that $|\mathbb{V}_{ij}| \leq 1$ w.l.o.g., we define $J : \Theta \times [-1, 1]^{N_h \times n} \to \mathbb{R}$ as

$$J(\boldsymbol{\mu}, \mathbb{V}) := \mathbb{E}_{\boldsymbol{\nu}}\| \boldsymbol{u}_{\boldsymbol{\mu},\boldsymbol{\nu}} - \mathbb{V}\mathbb{V}^T\mathbb{G}\boldsymbol{u}_{\boldsymbol{\mu},\boldsymbol{\nu}} \|, \tag{4.1}$$

and let some $\varepsilon > 0$. Then the continuity of J implies uniform continuity by the compactness of its pre-image, i.e. there is $\delta > 0$ such that

$$|\boldsymbol{\mu} - \boldsymbol{\mu}'| + \| \mathbb{V} - \mathbb{V}' \|_\infty < \delta \implies |J(\boldsymbol{\mu}, \mathbb{V}) - J(\boldsymbol{\mu}', \mathbb{V}')| < \varepsilon.$$

Then there is a Borel measurable map $s : \Theta \to [-1, 1]^{N_h \times n}$ given by

$$s : \boldsymbol{\mu} \mapsto \underset{\mathbb{V}\in[-1,1]^{N_h \times n}}{\operatorname{argmin}} J(\boldsymbol{\mu}, \mathbb{V}).$$

This map is bounded (given by continuity of J and compactness of its image) and hence $s \in L^1(\Theta; \mathbb{R}^{N_h \times n})$, i.e. $\| s \|_{L^1(\Theta;\mathbb{R}^{N_h \times n})} := \int_\Theta \| s(\boldsymbol{\mu}) \|_\infty d\mu(\boldsymbol{\mu}) < \infty$. This means that by Hornik's Theorem [Hor91] there is a deep ReLU neural network $\boldsymbol{V}_0 : \Theta \to \mathbb{R}^{N_h \times n}$ such that $\mathbb{E}_{\boldsymbol{\mu}}|\boldsymbol{V}_0(\boldsymbol{\mu}) -$

$s(\boldsymbol{\mu})| < \delta$ for all $\delta > 0$. Now let ρ be a entry-wise ReLU activation function and we consider a three-layered network $L : \mathbb{R}^{N_h \times n} \to \mathbb{R}^{N_h \times n}$ given by

$$L(\boldsymbol{x}) := \boldsymbol{e} - \mathbb{I}_{N_h \times n}\rho(-\mathbb{I}_{N_h \times n}\rho(\mathbb{I}_{N_h \times n}\boldsymbol{x} + \boldsymbol{e}) + 2\boldsymbol{e}),$$

where $\mathbb{I}_{N_h \times n}$ is the identity tensor on $\mathbb{R}^{N_h \times n}$, and $\boldsymbol{e}$ is a $N_h \times n$ matrix with ones as entries. Thus for each entry $x_{i,j}$ of $\boldsymbol{x}$, we have $L(x_{i,j}) = 1 - \rho(2 - \rho(x_{i,j} + 1)) \in [-1, 1]$. Thus, since L is 1-Lipschitz and L only effects values outside $[-1, 1]$, and hence $L \circ s = s$, we have for $\boldsymbol{V(\mu)} := L \circ \boldsymbol{V}_0(\boldsymbol{\mu})$

$$\mathbb{E}|\boldsymbol{V(\mu)} - s(\boldsymbol{\mu})| = \mathbb{E}|L \circ \boldsymbol{V}_0(\boldsymbol{\mu}) - L \circ s(\boldsymbol{\mu})| \leq \mathbb{E}|\boldsymbol{V}_0(\boldsymbol{\mu}) - s(\boldsymbol{\mu})| < \delta.$$

With this we can take $\boldsymbol{V(\mu)}$ to be in the pre-image of J and therefore conforming to its uniform continuity, i.e. take $\varepsilon > 0$, s.t.

$$|J(\boldsymbol{\mu}, s(\boldsymbol{\mu})) - J(\boldsymbol{\mu}, \boldsymbol{V}(\boldsymbol{\mu}))| < \varepsilon.$$

In total we calculate with $\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{\nu}}\| \boldsymbol{u}_{\boldsymbol{\mu},\boldsymbol{\nu}} - \boldsymbol{V}_{\boldsymbol{\mu}}\boldsymbol{V}_{\boldsymbol{\mu}}{}^T\mathbb{G}\boldsymbol{u}_{\boldsymbol{\mu},\boldsymbol{\nu}} \| = \mathbb{E}_{\boldsymbol{\mu}}[J(\boldsymbol{\mu}, \boldsymbol{V}(\boldsymbol{\mu})],$

$$\begin{aligned}\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{\nu}}\| \boldsymbol{u}_{\boldsymbol{\mu},\boldsymbol{\nu}} - \boldsymbol{V}_{\boldsymbol{\mu}}\boldsymbol{V}_{\boldsymbol{\mu}}{}^T\mathbb{G}\boldsymbol{u}_{\boldsymbol{\mu},\boldsymbol{\nu}} \| &\leq \mathbb{E}_{\boldsymbol{\mu}}[J(\boldsymbol{\mu}, \boldsymbol{V}(\boldsymbol{\mu})) - J(\boldsymbol{\mu}, s(\boldsymbol{\mu}))] + \mathbb{E}_{\boldsymbol{\mu}}[J(\boldsymbol{\mu}, s(\boldsymbol{\mu}))] \\ &< \varepsilon + \mathbb{E}_{\boldsymbol{\mu}}\left[\min_{\mathbb{V}} J(\boldsymbol{\mu}, \mathbb{V})\right] \\ &\leq \varepsilon + \mathbb{E}_{\boldsymbol{\mu}}\left[\min_{\mathbb{V}} \sup_{\boldsymbol{\nu}} \| \boldsymbol{u}_{\boldsymbol{\mu},\boldsymbol{\nu}} - \mathbb{V}\mathbb{V}^T\mathbb{G}\boldsymbol{u}_{\boldsymbol{\mu},\boldsymbol{\nu}} \|\right]\end{aligned}$$

□

Theorem 4.5. *Let $\mathcal{G} : (\mu, \nu, t) \mapsto u_h(\mu, \nu, t)$ for all $\mu \in \Theta, \nu \in \Theta'$ and $t \in \mathcal{T}$ be the parameter-to-solution map. Here we consider a general DOD+DNN structure, i.e. $\mathcal{G}(\mu, \nu, t) \approx V(\mu, t)\hat{q}(\mu, \nu, t)$, where $\hat{q} : \mathbb{R}^{p+q+1} \to \mathbb{R}^N$ is a neural network trained using data $(\mu_i, \nu_i, t_i)_{i=1}^{N_{data}}$ and $V : \mathbb{R}^{p+1} \to \mathbb{R}^{N_h \times N}$ is the DOD neural network, as described in (3.2.2). Then under the Assumptions (2) and (3), we have*

$$\mathcal{E}_R \leq \mathcal{E}_S + \mathcal{E}_{DOD} + \mathcal{E}_{NN}, \tag{4.2}$$

where

1. *The sampling error*
$$\mathcal{E}_S = \mathcal{E}_S\left(\mathcal{G}, (\mu_i, \nu_i, \tau_i)_{i=1}^{N_{data}}, N\right)$$
satisfies $\mathcal{E}_S \to 0$ as $N_s, N_t \to \infty$, with expectation
$$\mathbb{E}[\mathcal{E}_S] = \mathcal{O}(N_s^{-1/4} + N_t^{-1}).$$

2. *The DOD projection error*
$$\mathcal{E}_{DOD} = \mathcal{E}_{DOD}\left(\mathcal{G}, (\mu_i, \nu_i, \tau_i)_{i=1}^{N_{data}}, N\right)$$
satisfies $\mathcal{E}_{DOD} \to \mathcal{E}_{DOD,\infty}$ almost surely as $N_s, N_t \to \infty$, where
$$\mathcal{E}_{DOD,\infty} = \mathcal{E}_{DOD,\infty}(\mathcal{G}, N)$$

is independent of the data snapshots and arbitrarily close, with regards to the capabilities of the DOD neural network, to the expected linear KnW of the submanifold $\mathcal{S}_{\mu,t} = \{u(\mu,\nu,t) \,|\, \nu \in \Theta'\}$ *up to a constant, i.e.* $\mathbb{E}_{\mu,t}[d_N(\mathcal{S}_{\mu,t})]$.

3. *The neural network approximation error*

$$\mathcal{E}_{NN} = \mathcal{E}_{NN}(\mathcal{G}, N, \hat{q})$$

satisfies $\mathcal{E}_{NN} \to d_N(\mathcal{S}_\mu)$ *in expectation, as* $N_s, N_t \to \infty$.

Proof. In the core this proof is the same as the one of Theorem (4.2), we will therefore only consider the parts in which they are not interchangeable; the use of the triangle inequality gives us the neural network error in the same way, with only the additional factor being dependent on the choice and sophistication of the DOD map,

$$\mathcal{E}_{NN} = \left(\int_{\Theta\times\Theta'\times\mathcal{T}} \frac{\| V(\mu,t)q(\mu,\nu,t) - V(\mu,t)\hat{q}(\mu,\nu,t) \|^2}{\| u(\mu,\nu,t) \|^2} d(\mu,\nu,t) \right)^{1/2}. \tag{4.3}$$

Moreover, we can extrapolate the same procedure of splitting the remaining term into the respective errors

$$\| u(\mu,\nu,t) - V(\mu,t)V(\mu,t)^T u(\mu,\nu,t) \|_{L^2_\omega} \leq \mathcal{E}_S + \mathcal{E}_{\text{DOD}}, \tag{4.4}$$

where

$$\begin{aligned} \mathcal{E}_S = m^{-1} \Bigg| & \int_{\Theta\times\Theta'\times\mathcal{T}} \| u(\mu,\nu,t) - V(\mu,t)V(\mu,t)^T u(\mu,\nu,t) \|^2 d(\mu,\nu,t) \\ & - \frac{|\Theta\times\Theta'\times\mathcal{T}|}{N_{data}} \sum_{i=1}^{N_{s_1}} \sum_{j=1}^{N_{s_2}} \sum_{k=1}^{N_t} \| u(\mu_i,\nu_j,t_k) - V(\mu_i,t_k)V(\mu_i,t_k)^T u(\mu_i,\nu_j,t_k) \|^2 \Bigg|^{1/2} \end{aligned}$$

and

$$\mathcal{E}_{\text{DOD}} = \frac{m^{-1}|\Theta\times\Theta'\times\mathcal{T}|}{N_{data}} \Bigg| \sum_{i=1}^{N_{s_1}} \sum_{j=1}^{N_{s_2}} \sum_{k=1}^{N_t} \| u(\mu_i,\nu_j,t_k) - V(\mu_i,t_k)V(\mu_i,t_k)^T u(\mu_i,\nu_j,t_k) \|^2 \Bigg|^{1/2}.$$

This leaves us with the proof for the convergence statements. While $\mathcal{E}_S \to 0$ as $N_s, N_t \to \infty$ can be shown in the same manner as it has been in aforementioned proof above, the DOD error is more tricky.

Using the triangle inequality again and combining it with the findings of Lemma (4.4) we get for any $\varepsilon > 0$ and a suitable amount of complexity of the ReLU neural network $V : \mathbb{R}^{p+1} \to \mathbb{R}^{N_h \times N}$

$$\begin{aligned} \mathcal{E}_{\text{DOD}} &\leq m^{-1} \Bigg| \int_{\Theta\times\Theta'\times\mathcal{T}} \| u(\mu,\nu,t) - V(\mu,t)V(\mu,t)^T u(\mu,\nu,t) \|^2 d(\mu,\nu,t) \Bigg|^{1/2} \\ &\quad + \mathcal{E}_{\text{DOD}} - m^{-1} \Bigg| \int_{\Theta\times\Theta'\times\mathcal{T}} \| u(\mu,\nu,t) - V(\mu,t)V(\mu,t)^T u(\mu,\nu,t) \|^2 d(\mu,\nu,t) \Bigg|^{1/2} \\ \underset{N_s, N_t \to \infty}{\longrightarrow} & m^{-1} \Bigg| \int_{\Theta\times\Theta'\times\mathcal{T}} \| u(\mu,\nu,t) - V(\mu,t)V(\mu,t)^T u(\mu,\nu,t) \|^2 d(\mu,\nu,t) \Bigg|^{1/2} \\ &< m^{-1}\varepsilon + m^{-1} \int_{\Theta\times\mathcal{T}} d_N(\mathcal{S}_{\mu,t}) d(\mu,t) \end{aligned}$$

□

Lemma 4.6 (Projection Error)**.** Let N_{s_2} be given under Assumption (2), then for all $\varepsilon > 0$ there is a ReLU neural network $V : \mathbb{R}^{p+1} \to \mathbb{R}^{N_h \times N}$, such that for $u_j := u(\mu, \nu_j, t)$ with given $\mu \in \Theta$ and $t \in \mathcal{T}$

$$\sup_{(\mu,t)\in\Theta\times\mathcal{T}} \left| \frac{1}{N_{s_2}} \sum_{j=1}^{N_{s_2}} \| u_j - V(\mu,t)V(\mu,t)^T u_j \|^2 - \min_{W\in\mathbb{R}^{N_h\times N}} \frac{1}{N_{s_2}} \sum_{j=1}^{N_{s_2}} \| u_j - WW^T u_j \|^2 \right|^{1/2} < \varepsilon.$$

Proof. The existence and identity of V^* s.t.

$$V^* = \operatorname*{argmin}_{W\in\mathbb{R}^{N_h\times N}} \sum_{j=1}^{N_{s_2}} \| u_j - V^*(V^*)^T u_j \|^2$$

is given via the statement in Proposition (2.8) with the POD basis. Hence defining a norm via

$$\| (u_j)_{j=1}^{N_{s_2}} \|_{train} := \left| \sum_{j=1}^{N_{s_2}} \| u_j \|^2 \right|^{1/2}$$

gives us the following inequality firstly using the monotonicity of the square root and then using the triangle inequalities multiple times

$$\begin{aligned} &N_{s_2}^{-1/2} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \left| \sum_{j=1}^{N_{s_2}} \| u_j - V(\mu,t)V(\mu,t)^T u_j \|^2 - \sum_{j=1}^{N_{s_2}} \| u_j - V^*(V^*)^T u_j \|^2 \right|^{1/2} \\ \leq\, &N_{s_2}^{-1/2} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \left| \sum_{j=1}^{N_{s_2}} \| V(\mu,t)V(\mu,t)^T u_j - V^*(V^*)^T u_j \|^2 \right|^{1/2}, \end{aligned}$$

then define $q_j := (V^*)^T u_j$ and $\hat{q}_j := V(\mu,t)^T u_j$ and their training vectors accordingly, i.e. $q := (q_j)_{j=1}^{N_{s_2}}, \hat{q} := (\hat{q}_j)_{j=1}^{N_{s_2}}$

$$\begin{aligned} &\leq N_{s_2}^{-1/2} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \| V(\mu,t)\hat{q} - V(\mu,t)q \|_{train} + \| V(\mu,t)q - V^* q \|_{train} \\ &\leq N_{s_2}^{-1/2} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \| V(\mu,t) \|_\infty \| \hat{q} - q \|_{train} + \| V(\mu,t) - V^* \|_\infty \| q \|_{train} \\ &= N_{s_2}^{-1/2} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \| V(\mu,t) \|_\infty \| V(\mu,t)^T u - (V^*)^T u \|_{train} + \| V(\mu,t) - V^* \|_\infty \| (V^*)^T u \|_{train} \\ &\leq N_{s_2}^{-1/2} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \| V(\mu,t) \|_\infty \| V(\mu,t) - V^* \|_\infty \| u \|_{train} + \| V(\mu,t) - V^* \|_\infty \| V^* \|_\infty \| u \|_{train} \\ &\leq N_{s_2}^{-1/2} \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \| u \|_{train} (\| V(\mu,t) - V^* \|_\infty (\| V(\mu,t) \|_\infty + 1)) \\ &\leq 2M \sup_{(\mu,t)\in\Theta\times\mathcal{T}} \| V(\mu,t) - V^* \|_\infty, \end{aligned}$$

where we used $\| u \|_{train} \leq | \sum_{j=1}^{N_{s_2}} M^2 |^{1/2} = N_{s_2}^{1/2} M$ and additionally $\| V(\mu,t) \|_\infty = \| V^* \|_\infty = 1$ by orthonormality in the last arguments.

Do the Yarotski argument with the appropriate number of layers and weights.

□

4.3 Complexity Bounds for POD-DL-ROMs

Include discussed results with proof of Brivio et al.

Theorem 4.7. *Let $\mathcal{G} : (\mu, \nu, t) \mapsto u(\mu, \nu, t)$ for any $(\mu, \nu, t) \in \Theta \times \Theta' \times \mathcal{T}$ be the parameter-to-solution map and let Assumption (2) and (3) be true. Let $1 > \delta > 0$ and $\varepsilon > 0$. Now assume we can collect freely a sufficient amount of training data $N_{data} = N_{data}(\delta, \varepsilon)$ and put it into the correlation matrix K as given in Definition (2.6) with σ_k^2 being its sorted eigenvalues. Now choose*

$$N = \operatorname{argmin} \left\{ j \in \mathbb{N} \,\middle|\, \sum_{k>j} \sigma_k^2 \leq \frac{m^2}{9} \varepsilon^2 \right\}.$$

Given the POD matrix $V \in \mathbb{R}^{N_h \times N}$ we now define the parameter-to-POD-coefficient map $\mathcal{Q} : (\mu, \nu, t) \mapsto q(\mu, \nu, t) := V^T u(\mu, \nu, t)$ for any $(\mu, \nu, t) \in \Theta \times \Theta' \times \mathcal{T}$. We assume that there exists $n > 0, \psi_* : \mathbb{R}^n \to \mathbb{R}^N, \psi'_* : \mathbb{R}^N \to \mathbb{R}^n$ that are respectively s-times and s'-times differentiable (with $s >> s' \geq 2$), such that they enjoy the perfect embedding Assumption (4), i.e.

$$\psi_*(\psi'_*(q(\mu, \nu, t))) = q(\mu, \nu, t), \quad \forall (\mu, \nu, t) \in \Theta \times \Theta' \times \mathcal{T}.$$

We further let

$$C_1 := \sup_{|\alpha| \leq s'} \sup_{v \in \mathbb{R}^N} |\, D^\alpha \psi'_*(v) \,|, \quad C_2 := \sup_{|\alpha| \leq s} \sup_{w \in \mathbb{R}^n} |\, D^\alpha \psi'_*(w) \,|.$$

Then there is a constant $c = c(\Theta, \Theta', \mathcal{T}, L, C_1, C_2, p, q, n, s, s')$ and a POD-DL-ROM architecture $V\hat{q} = V\psi \circ \phi : \mathbb{R}^{p+q+1} \to \mathbb{R}^{N_h}$ composed of a decoder $\psi : \mathbb{R}^n \to \mathbb{R}^N$ having at most:

- *$L_{n \to N} = c \log(\varepsilon^{-1})$ layers,*
- *$\omega_{n \to N} = cN\varepsilon^{-n/(s-1)} \log(\varepsilon^{-1})$ active weights,*

and a reduced feed forward neural network $\phi : \mathbb{R}^{p+q+1} \to \mathbb{R}^n$ having at most:

- *$L_{(p+q+1) \to n} = c \log(\varepsilon^{-1})$ layers,*
- *$\omega_{(p+q+1) \to n} = cn\varepsilon^{-(p+q+1)} \log(\varepsilon^{-1})$ active weights,*

such that $\mathbb{P}[\mathcal{E}_R < \varepsilon] > 1 - \delta$.

Proof. We can directly bound $\mathcal{E}_S = \mathcal{E}_S(N_{s_1}, N_{s_2}, N_t)$ independently of N, under the Assumption (2) by the Weak Law of Large Numbers, i.e. for all $1 > \delta > 0$ and $\varepsilon > 0$ there are N_{s_1}, N_{s_2}, N_t, such that

$$\mathbb{P}[\mathcal{E}_S((N_{s_1}, N_{s_2}, N_t)) < \varepsilon/3] > 1 - \delta. \tag{4.1}$$

This result provides the "probability" part of the PAC (probably almost correct) bound, which we want to show. Bounding $\mathcal{E}_{\text{NN}}$ and $\mathcal{E}_{\text{POD}}$ by $\varepsilon/3$ respectively will then provide the wanted result. For this, by choosing N as in the assumption of the statement, we simply rewrite

$$\mathcal{E}_{\text{POD}} := m^{-1} \sqrt{\sum_{k>N} \sigma_k^2} \leq \frac{\varepsilon}{3}. \tag{4.2}$$

To now bound the neural network approximation error, we will need to simply rewrite the error in terms of only the norm of the network itself;

$$
\begin{aligned}
\mathcal{E}_{\mathrm{NN}} &= \left(\int_{\Theta\times\Theta'\times\mathcal{T}} \frac{\| Vq(\mu,\nu,t) - V\hat{q}(\mu,\nu,t)\|^2}{\| u(\mu,\nu,t)\|^2}\, d(\mu,\nu,t)\right)^{1/2} \\
&\leq m^{-1}\left(\int_{\Theta\times\Theta'\times\mathcal{T}} \| Vq(\mu,\nu,t) - V\hat{q}(\mu,\nu,t)\|^2\, d(\mu,\nu,t)\right)^{1/2} \\
&\leq m^{-1}\left(|\Theta\times\Theta'\times\mathcal{T}| \sup_{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}} \| Vq(\mu,\nu,t) - V\hat{q}(\mu,\nu,t)\|^2\right)^{1/2} \\
&\leq m^{-1}\left(|\Theta\times\Theta'\times\mathcal{T}|\, \| V\|_\infty^2 \sup_{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}} \| q(\mu,\nu,t) - \hat{q}(\mu,\nu,t)\|^2\right)^{1/2} \\
&= m^{-1}\, |\Theta\times\Theta'\times\mathcal{T}|^{1/2} \sup_{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}} \| q(\mu,\nu,t) - \hat{q}(\mu,\nu,t)\|.
\end{aligned}
\tag{4.3}
$$

whereas we split the latter error into both parts, since $\hat{q} = \psi\circ\phi$ per definition. This means employing two types of error estimations in the matter of finding the respective optimal maps for the feed forward estimation of the latent dynamics via $\phi_* : \mathbb{R}^{(p+q+1)} \to \mathbb{R}^n$ and the decoder lift via $\psi_* : \mathbb{R}^n \to \mathbb{R}^N$.

- Let $\mathcal{S}_N := \{q = \mathcal{Q}(\mu,\nu,t) \,|\, (\mu,\nu,t) \in \Theta\times\Theta'\times\mathcal{T}\}$, then the image of the optimal embedding $\mathcal{V}_n = \psi'_*(\mathcal{S}_N)$ is such that $\mathrm{diam}(\mathcal{V}_N) \leq LC_1\mathrm{diam}(\Theta\times\Theta'\times\mathcal{T})$, given the Assumption (3) by continuity.

  Make this rigorous

  Thus, by Theorem Gühring, there is a ReLU Neural Network $\psi : \mathbb{R}^n \to \mathbb{R}^N$, such that

$$
\sup_{v\in\mathcal{V}_N} \| \psi(v) - \psi_*(v)\| < \frac{m}{6}|\,\Theta\times\Theta'\times\mathcal{T}\,|^{-1/2}\varepsilon, \tag{4.4}
$$

$$
\operatorname*{ess\,sup}_{v,v'\in\mathcal{V}_N} \frac{|\,(\psi-\psi_*)(v) - (\psi-\psi_*)(v')\,|}{|\,v - v'\,|} < \frac{m}{6}|\,\Theta\times\Theta'\times\mathcal{T}\,|^{-1/2}\varepsilon, \tag{4.5}
$$

  with $L_{n\to N} = c\log(\varepsilon^{-1})$ layers and $\omega_{n\to N} = cN\varepsilon^{-n/(s-1)}\log(\varepsilon^{-1})$ active weights.

  Make this rigorous

- Now let $\phi_*(\mu,\nu,t) = \psi'_*(q(\mu,\nu,t))$ for $(\mu,\nu,t) \in \Theta\times\Theta'\times\mathcal{T}$, which is Lipschitz continuous, with constant LC_1, and thus, by

  Make this rigorous

  Theorem Yarotski, there exists a ReLU Neural Network $\phi : \mathbb{R}^{p+q+1} \to \mathbb{R}^n$, such that

$$
\sup_{(\mu,\nu,t)\in\Theta\times\Theta\times\mathcal{T}} \| \phi(\mu,\nu,t) - \phi_*(\mu,\nu,t)\| < \frac{m}{6C_3}|\,\Theta\times\Theta'\times\mathcal{T}\,|^{-1/2}\varepsilon, \tag{4.6}
$$

  with $L_{(p+q+1)\to n} = c\log(\varepsilon^{-1})$ layers and $\omega_{(p+q+1)\to n} = cn\varepsilon^{-(p+q+1)}\log(\varepsilon^{-1})$ active weights.

Starting from the last inequality in (4.3) we can perform the split in a clean fashion using the triangle inequality and reminding ourselves of the Assumption (4), which guarantees $q = \psi_*(\psi'_*)$, and thus gives with both neural network estimates in Equations (4.4) and (4.6)

$$
\begin{aligned}
&\sup_{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}} \| q(\mu,\nu,t) - \hat{q}(\mu,\nu,t) \| \\
&\le \sup_{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}} \left(\| \psi_*(\psi'_*(\mu,\nu,t)) - \psi(\phi_*(\mu,\nu,t)) \| + \| \psi(\phi_*(\mu,\nu,t)) - \psi(\phi(\mu,\nu,t)) \| \right) \\
&\le \sup_{v\in\mathcal{V}_N} \| \psi_*(v) - \psi(v) \| + C_3 \sup_{(\mu,\nu,t)\in\Theta\times\Theta'\times\mathcal{T}} \left(\| \psi(\phi_*(\mu,\nu,t)) - \psi(\phi(\mu,\nu,t)) \| \right) \\
&< \frac{m}{6} | \Theta\times\Theta'\times\mathcal{T} |^{-1/2}\varepsilon + C_3 \frac{m}{6C_3} | \Theta\times\Theta'\times\mathcal{T} |^{-1/2}\varepsilon = \frac{\varepsilon}{3} m \, | \Theta\times\Theta'\times\mathcal{T} |^{-1/2},
\end{aligned}
$$

which cancels with $m^{-1}\,|\Theta\times\Theta'\times\mathcal{T}|^{1/2}$ to get

$$
\mathcal{E}_{\mathrm{NN}} < \frac{\varepsilon}{3}. \tag{4.7}
$$

Now putting together all three estimates (4.1), (4.2) and (4.7) with the reasoning of Theorem (4.2) finally gives

$$
\mathcal{E}_R \le \mathcal{E}_S + \mathcal{E}_{\mathrm{POD}} + \mathcal{E}_{\mathrm{NN}} < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon,
$$

with higher probability than $1-\delta$. □

4.4 Complexity Bounds for DOD-DL-ROMs

Extrapolate to DOD-DL-ROM

Theorem 4.8. *Let $\mathcal{G} : (\mu,\nu,t) \mapsto u(\mu,\nu,t)$ for any $(\mu,\nu,t) \in \Theta\times\Theta'\times\mathcal{T}$ be the parameter-to-solution map and let Assumption (2) and (3) be true. Let $1 > \delta > 0$ and $\varepsilon > 0$. Now assume we can collect freely a sufficient amount of training data Θ_{data}, Θ'_{data} and $\mathcal{T}_{data}$ of size $N_{data} = N_{data}(\delta,\varepsilon)$ and put all samples of $u(\mu,\nu_j,t)_{j=1}^{N_{s_2}}$ into a correlation matrix $K_{\mu,t}$ for each sample $(\mu,t) \in \Theta_{data}\times\mathcal{T}_{data}$ as given in Definition (2.6) with $\sigma_k(\mu,t)^2_k$ being its sorted eigenvalues. Now choose*

$$
N_{DOD} = \operatorname{argmin}\left\{ j\in\mathbb{N} \,\middle|\, \sup_{(\mu,t)\in\Theta_{data}\times\mathcal{T}_{data}} \sum_{k>j} \sigma(\mu,t)^2_k \le \frac{m^2}{16}\varepsilon^2 \right\}.
$$

For each pair $(\mu,t) \in \Theta_{data}\times\mathcal{T}_{data}$ given the respective POD matrix, so the best possible projection given the data set (refer to (2.8)), $V^*(\mu,t) \in \mathbb{R}^{N_h\times N_{DOD}}$ we now define the parameter-to-optimal-coefficient map $\mathcal{Q} : (\mu,\nu,t) \mapsto q(\mu,\nu,t) := V^*(\mu,t)^T u(\mu,\nu,t)$ for any $(\mu,\nu,t) \in \Theta_{data}\times\Theta'\times\mathcal{T}_{data}$. We assume that there exists $n > 0, \Psi_* : \mathbb{R}^n \to \mathbb{R}^{N_{DOD}}, \Psi'_* : \mathbb{R}^{N_{DOD}} \to \mathbb{R}^n$ that are respectively s-times and s'-times differentiable (with $s >> s' \ge 2$), such that they enjoy the perfect embedding Assumption (4), i.e.

$$
\Psi_*(\Psi'_*(q(\mu,\nu,t))) = q(\mu,\nu,t), \quad \forall (\mu,\nu,t) \in \Theta\times\Theta'\times\mathcal{T}.
$$

We further let

$$
C'_1 := \sup_{|\alpha|\le s'} \sup_{v\in\mathbb{R}^{N_{DOD}}} | D^\alpha \Psi'_*(v) |, \quad C'_2 := \sup_{|\alpha|\le s} \sup_{w\in\mathbb{R}^n} | D^\alpha \Psi'_*(w) |.
$$

Then there is a constant $c = c(\Theta, \Theta', \mathcal{T}, L, C_1, C_2, p, q, n, s, s')$ and a DOD-DL-ROM architecture $V\hat{q} = V_{DOD}\Psi \circ \phi : \mathbb{R}^{p+q+1} \to \mathbb{R}^{N_h}$ composed of a decoder $\Psi : \mathbb{R}^n \to \mathbb{R}^{N_{DOD}}$ having at most:

- $L_{n \to N_{DOD}} = c \log(\varepsilon^{-1})$ *layers,*
- $\omega_{n \to N_{DOD}} = c\, N_{DOD}\, \varepsilon^{-n/(s-1)} \log(\varepsilon^{-1})$ *active weights,*

a reduced feed forward neural network $\phi : \mathbb{R}^{p+q+1} \to \mathbb{R}^n$ having at most:

- $L_{(p+q+1) \to n} = c \log(\varepsilon^{-1})$ *layers,*
- $\omega_{(p+q+1) \to n} = cn\varepsilon^{-(p+q+1)} \log(\varepsilon^{-1})$ *active weights,*

and a DOD Module $V_{DOD} : \mathbb{R}^{p+1} \to \mathbb{R}^{N_h \times N_{DOD}}$ having at most:

- $L_{(p+1) \to N_h \times N_{DOD}} = c \log(\varepsilon^{-1})$ *layers,*
- $\omega_{(p+1) \to N_h \times N_{DOD}} = c\, N_h\, N_{DOD}\, \varepsilon^{-(p+1)} \log(\varepsilon^{-1})$ *active weights,*

such that $\mathbb{P}[\mathcal{E}_R < \varepsilon] > 1 - \delta$.

Proof. Similar as to the proof of Theorem (4.7), we will try to bound each quantity of the statement in Theorem (4.5), but with the catch, that $\mathcal{E}_{\text{DOD}}$ is stochastic in nature, and thus we will employ Lemma (4.6) to find an auxilary term, which is independent of the data, to which the approximation of the neural net of the DOD is sufficiently "close".

The sampling error can be bounded in the same way as before, i.e. we can directly bound $\mathcal{E}_S = \mathcal{E}_S(N_{s_1}, N_{s_2}, N_t)$ independently of N_{DOD}, under the Assumption (2) by the Weak Law of Large Numbers, i.e. for all $1 > \delta > 0$ and $\varepsilon > 0$ there are N_{s_1}, N_{s_2}, N_t, such that

$$\mathbb{P}[\mathcal{E}_S((N_{s_1}, N_{s_2}, N_t)) < \varepsilon/4] > 1 - \delta. \tag{4.1}$$

Now recall the definition of $\mathcal{E}_{\text{DOD}}$ in the proof of Theorem (4.5) and further establish

$$\begin{aligned}\mathcal{E}_{\text{DOD}} &= \frac{m^{-1}|\Theta \times \Theta' \times \mathcal{T}|}{N_{data}^{1/2}} \left| \sum_{i=1}^{N_{s_1}} \sum_{j=1}^{N_{s_2}} \sum_{k=1}^{N_t} \| u(\mu_i, \nu_j, t_k) - V(\mu_i, t_k) V(\mu_i, t_k)^T u(\mu_i, \nu_j, t_k) \|^2 \right|^{1/2} \\ &\leq m^{-1} \sup_{(\mu,t) \in \Theta \times \mathcal{T}} \left| \frac{|\Theta \times \Theta' \times \mathcal{T}|}{N_{s_2}} \sum_{j=1}^{N_{s_2}} \| u(\mu, \nu_j, t) - V(\mu, t) V(\mu, t)^T u(\mu, \nu_j, t) \|^2 \right|^{1/2},\end{aligned}$$

enabling us to use Proposition (2.8) in order to switch to the eigenvalues $\sigma(\mu, t)_k$ for $(\mu, t) \in \Theta_{\text{data}} \times \mathcal{T}_{\text{data}}$ of the theoretical POD matrix $V^*(\mu, t)$ by employing the Lemma (4.6) with the bound

$$\frac{\varepsilon\, m^2}{4|\Theta \times \Theta' \times \mathcal{T}|^{1/2}},$$

which directly gives us the approximation result via the defition of N_{DOD}

$$\mathcal{E}_{\text{DOD}} < \frac{\varepsilon}{4} + m^{-1} \sup_{(\mu,t) \in (\Theta_{\text{data}} \times \mathcal{T}_{\text{data}})} \sqrt{\sum_{k > N_{\text{DOD}}} \sigma(\mu, t)_k^2} \leq \frac{\varepsilon}{2}. \tag{4.2}$$

Include the cost via Yarotski

Argue the actual part for the NN error

Moreover we can use the exact same arguments as in the bound for $\mathcal{E}_{\text{NN}}$ for the POD-DL-ROM as in the proof of Theorem (4.7), which yields

$$\mathcal{E}_{\text{NN}} < \frac{\varepsilon}{4}. \tag{4.3}$$

Finally putting each Equation, i.e. (4.1), (4.2) and (4.3), together, we get

$$\mathcal{E}_R \leq \mathcal{E}_S + \mathcal{E}_{\text{DOD}} + \mathcal{E}_{\text{NN}} < \frac{\varepsilon}{4} + \frac{\varepsilon}{2} + \frac{\varepsilon}{4} = \varepsilon,$$

with probability greater than $1 - \delta$ concluding our proof. $\square$

4.5 Comparison between existing Techniques

Chapter 5

Numerical Experiments

Include all numerical experiments

Find increase of performance depending on latent dimension

Give benchmark on pass forward times

Discuss comparison between different DL-ROMs

Chapter 6

Conclusion

Discuss results

Insert outlook for future ideas and implementation

Bibliography

[Alt16] Hans Wilhelm Alt. *Linear Functional Analysis*. Springer London, July 2016. ISBN: 978-1-4471-7279-6, 978-1-4471-7280-2. DOI: `10.1007/978-1-4471-7280-2`.

[BB98] Haïm Brezis and Felix Browder. "Partial Differential Equations in the 20th Century". In: *Advances in Mathematics* 135.1 (1998), pp. 76–144. ISSN: 0001-8708. DOI: `https://doi.org/10.1006/aima.1997.1713`. URL: `https://www.sciencedirect.com/science/article/pii/S0001870897917138`.

[BH70] J. H. Bramble and S. R. Hilbert. "Bounds for a Class of Linear Functionals with Applications to Hermite Interpolation". In: *Numerische Mathematik* 16.4 (1970), pp. 362–369. DOI: `10.1007/BF02165086`. URL: `http://eudml.org/doc/132041`.

[Dac04] Bernard Dacorogna. *Introduction to the Calculus of Variations*. Nov. 2004. ISBN: 978-1-86094-499-4. DOI: `10.1142/p361`.

[FDM20] Stefania Fresca, Luca Dede, and Andrea Manzoni. *A comprehensive deep learning-based approach to reduced order modeling of nonlinear time-dependent parametrized PDEs*. 2020. arXiv: `2001.04001 [math.NA]`. URL: `https://arxiv.org/abs/2001.04001`.

[Fra+24] Nicola Rares Franco et al. *Deep orthogonal decomposition: a continuously adaptive data-driven approach to model order reduction*. Preprint, arXiv:2404.18841 [math.NA] (2024). arXiv:2404.18841. 2024. URL: `https://arxiv.org/abs/2404.18841`.

[GKP19] Ingo Gühring, Gitta Kutyniok, and Philipp Petersen. *Error bounds for approximations with deep ReLU neural networks in $W^{s,p}$ norms*. 2019. arXiv: `1902.07896 [math.FA]`. URL: `https://arxiv.org/abs/1902.07896`.

[Hor91] Kurt Hornik. "Approximation capabilities of multilayer feedforward networks". In: *Neural Networks* 4.2 (1991), pp. 251–257. ISSN: 0893-6080. DOI: `https://doi.org/10.1016/0893-6080(91)90009-T`. URL: `https://www.sciencedirect.com/science/article/pii/089360809190009T`.

[QMN15] Alfio Quarteroni, Andrea Manzoni, and Federico Negri. *Reduced Basis Methods for Partial Differential Equations*. Springer, 2015. DOI: `10.1007/978-3-319-15431-2`.

[Yar17] Dmitry Yarotsky. "Error bounds for approximations with deep ReLU networks". In: *Neural Networks* 94 (2017), pp. 103–114. ISSN: 0893-6080. DOI: `https://doi.org/10.1016/j.neunet.2017.07.002`. URL: `https://www.sciencedirect.com/science/article/pii/S0893608017301545`.